

NCERT Exemplar Solutions For Grade 12

Mathematics

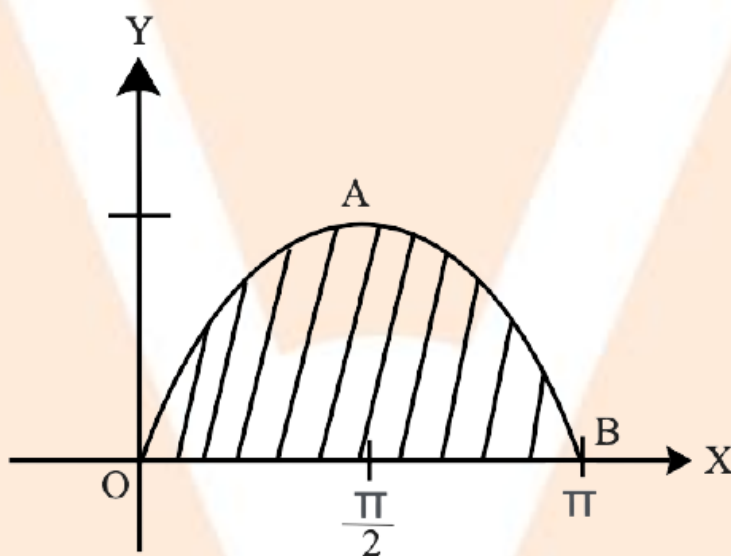
Chapter 8: Application of Integrals

Solved Examples

Short Answers

1. Find the area of the curve $y = \sin x$ between 0 and π .

Ans: We have,



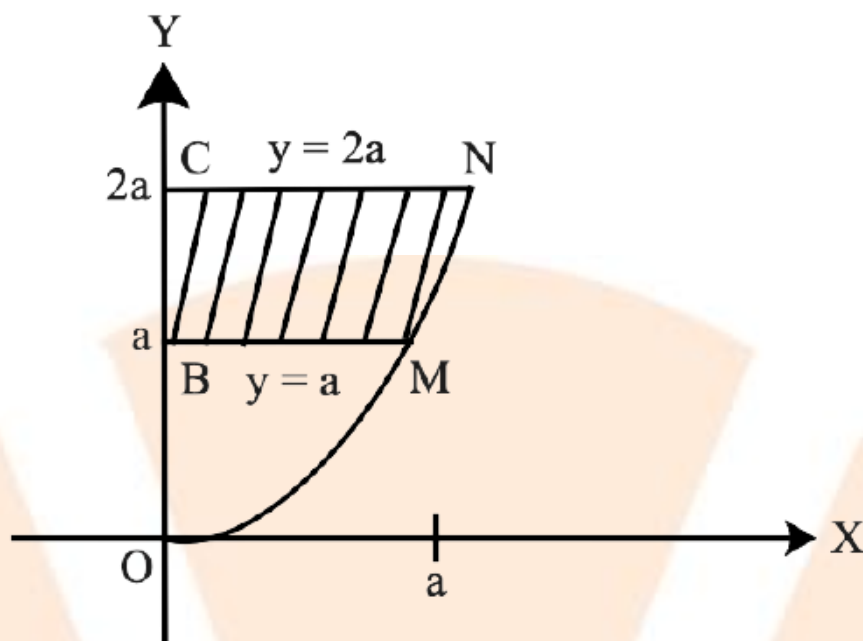
$$\text{Area OAB} = \int_0^{\pi} y dx = \int_0^{\pi} \sin x dx$$

$$= \left| -\cos x \right|_0^{\pi}$$

$$= \cos 0 - \cos \pi = 2 \text{ sq units.}$$

2. Find the area of the region bounded by the curve $ay^2 = x^3$, the y-axis and the lines $y = a$ and $y = 2a$.

Ans: We have



$$\text{Area BMNC} = \int_a^{2a} x dy = \int_a^{2a} a^{\frac{1}{3}} y^{\frac{2}{3}} dy$$

$$= \frac{3a^{\frac{1}{3}}}{5} \left| y^{\frac{5}{3}} \right|_a^{2a}$$

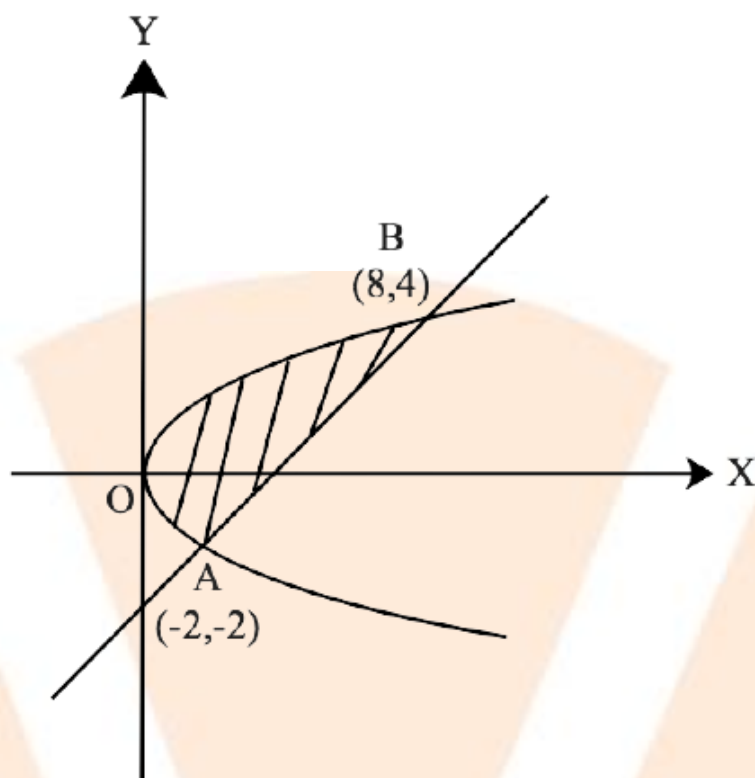
$$= \frac{3a^{\frac{1}{3}}}{5} \left| (2a)^{\frac{5}{3}} - a^{\frac{5}{3}} \right|$$

$$= \frac{3a^{\frac{1}{3}} a^{\frac{5}{3}}}{5} \left| 2^{\frac{5}{3}} - 1 \right|$$

$$= \frac{3a^2}{5} \left| 2 \cdot 2^{\frac{1}{3}} - 1 \right| \text{ sq. units}$$

3. Find the area of the region bounded by the parabola $y^2 = 2x$ and the straight line $x - y = 4$.

Ans:

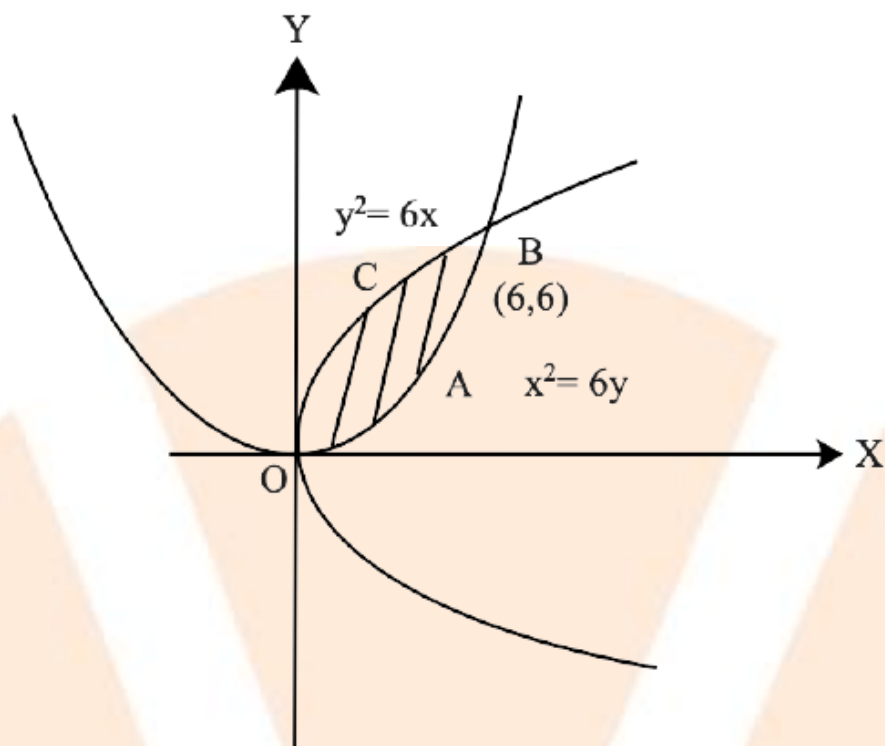


The intersecting points of the given curves are obtained by solving the equations $x - y = 4$ and $y^2 = 2x$ for x and y . We have $y^2 = 8 + 2y$ i.e., $(y - 4)(y + 2) = 0$ which gives $y = 4, -2$ and $x = 8, 2$. Thus, the points of intersection are $(8, 4), (2, -2)$. Hence

$$\begin{aligned} \text{Area} &= \int_{-2}^4 \left(4 + y - \frac{y^2}{2} \right) dy \\ &= \left[4y + \frac{y^2}{2} - \frac{y^3}{6} \right]_{-2}^4 = 18 \text{ sq. units} \end{aligned}$$

4. Find the area of the region bounded by the parabolas $y^2 = 6x$ and $x^2 = 6y$.

Ans:



The intersecting points of the given parabolas are obtained by solving these equations for x and y , which are $0(0, 0)$ and $(6, 6)$. Hence

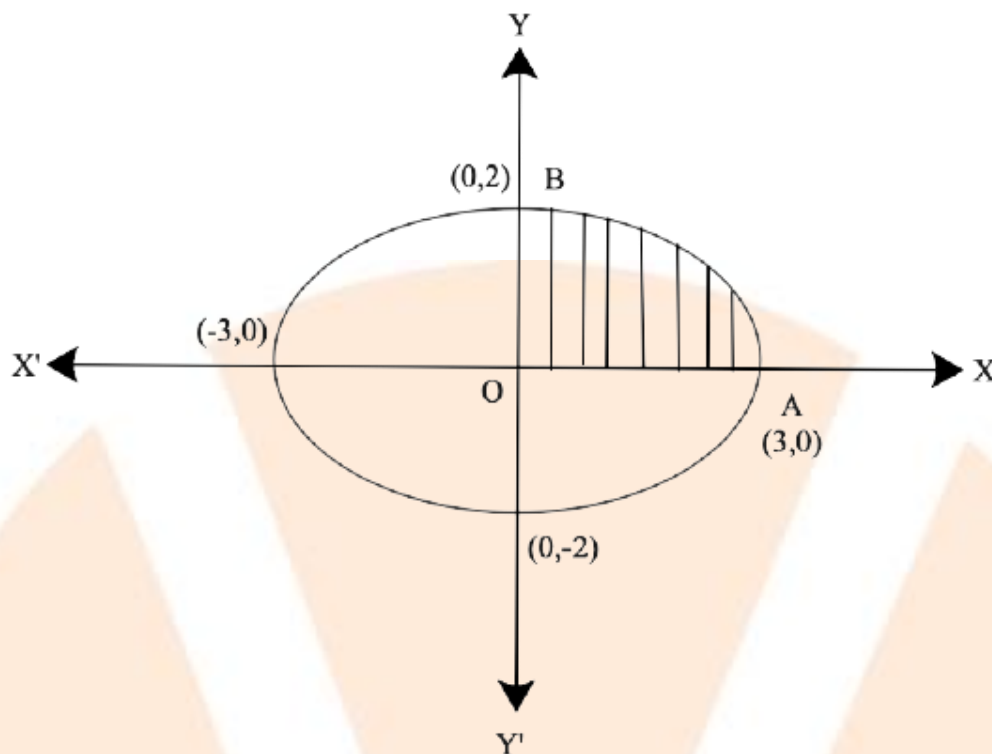
$$\begin{aligned} \text{Area OABC} &= \int_0^6 \left(\sqrt{6x} - \frac{x^2}{6} \right) dx = \left| 2\sqrt{6} \frac{x^{3/2}}{3} - \frac{x^3}{18} \right|_0^6 \\ &= \left| 2\sqrt{6} \frac{6^{3/2}}{3} - \frac{6^3}{18} \right| = 12 \text{ sq. units} \end{aligned}$$

5. Find the area enclosed by the curve $x = 3 \cos t$, $y = 2 \sin t$.

Ans: Eliminating t as follows: $x = 3 \cos t$, $y = 2 \sin t \Rightarrow \cos t = \frac{x}{3}$, $\sin t = \frac{y}{2}$, we obtain

$$\frac{x^2}{9} + \frac{y^2}{4} = 1,$$

Which is the equation of an ellipse.



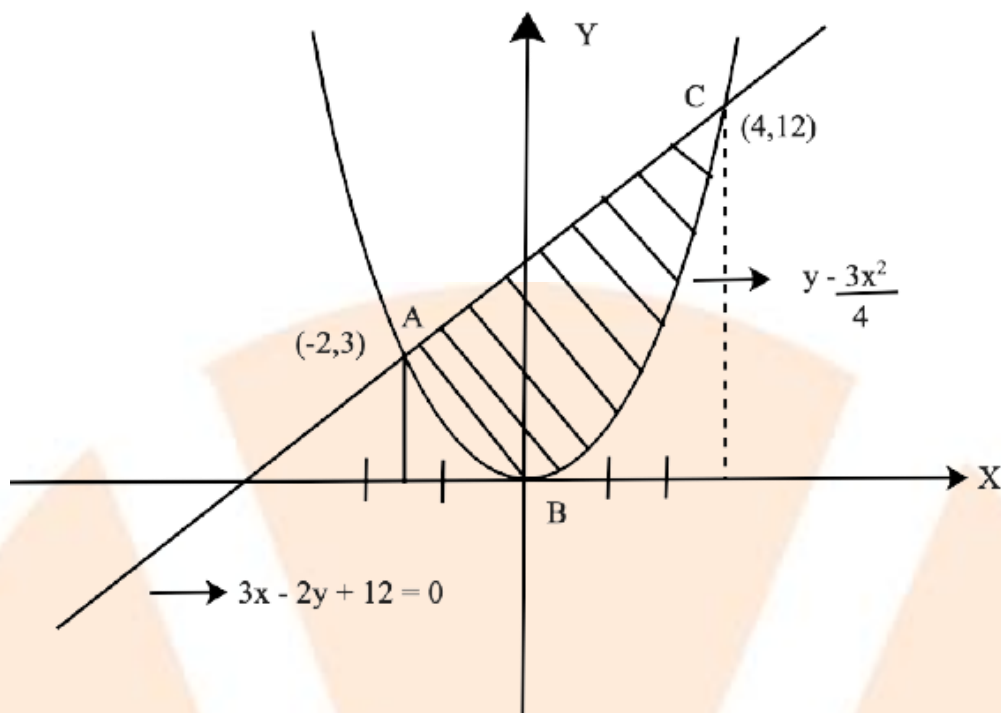
From figure we get:

$$\begin{aligned} \text{The required area} &= 4 \int_0^3 \frac{2}{3} \sqrt{9-x^2} dx \\ &= \frac{8}{3} \left[\frac{x}{2} \sqrt{9-x^2} + \frac{9}{2} \sin^{-1} \frac{x}{3} \right]_0^3 = 6\pi \text{ sq. units} \end{aligned}$$

Long Answers

6. Find the area of the region included between the parabola $y = \frac{3x^2}{4}$ and the line $3x - 2y + 12 = 0$.

Ans: Solving the equations of the given curves $y = \frac{3x^2}{4}$ and $3x - 2y + 12 = 0$, we get
 $3x^2 - 6x - 24 = 0 \Rightarrow (x-4)(x+2) = 0$
 $\Rightarrow x = 4, x = -2$ which give $y = 12, y = 3$



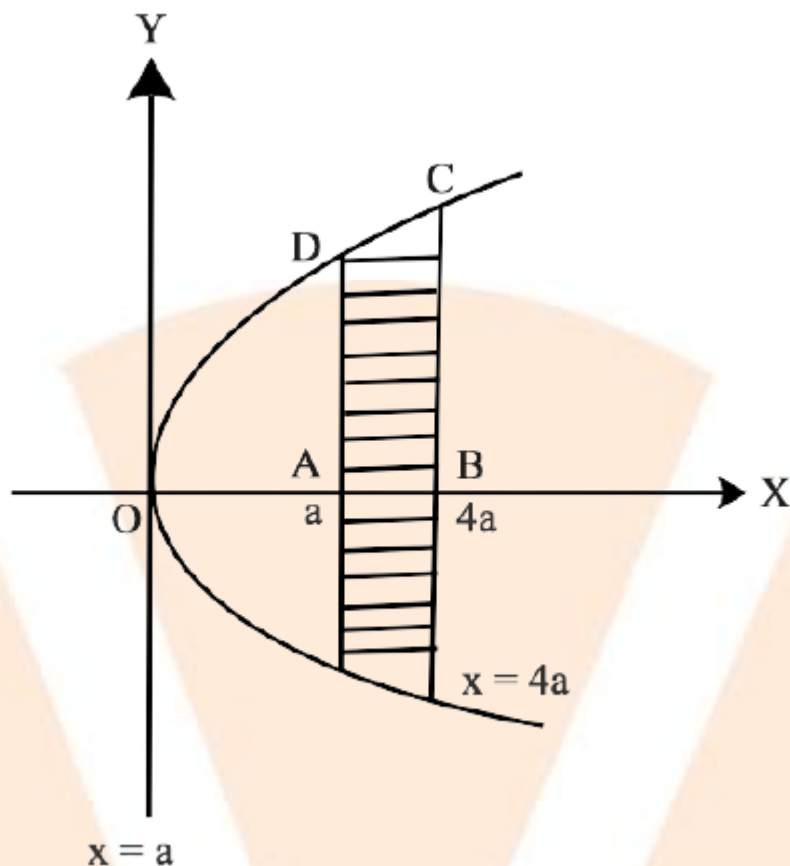
From Fig, the required area = area of ABC

$$\begin{aligned}
 &= \int_{-2}^4 \left(\frac{12+3x}{2} \right) dx - \int_{-2}^4 \left(\frac{3x^2}{4} \right) dx \\
 &= \left(6x + \frac{3x^2}{4} \right)_{-2}^4 - \left| \frac{3x^3}{12} \right|_{-2}^4 = 27 \text{ sq. units}
 \end{aligned}$$

7. Find the area of the region bounded by the curves $x = at^2$ and $y = 2at$ between the ordinates corresponding to $t = 1$ and $t = 2$.

Ans: Given that $x = at^2$...(i), $y = 2at$...(ii) $\Rightarrow t = \frac{y}{2a}$ putting the value of t in (i), we get $y^2 = 4ax$

Putting $t = 1$ and $t = 2$ in (i), we get $x = a$, and $x = 4a$



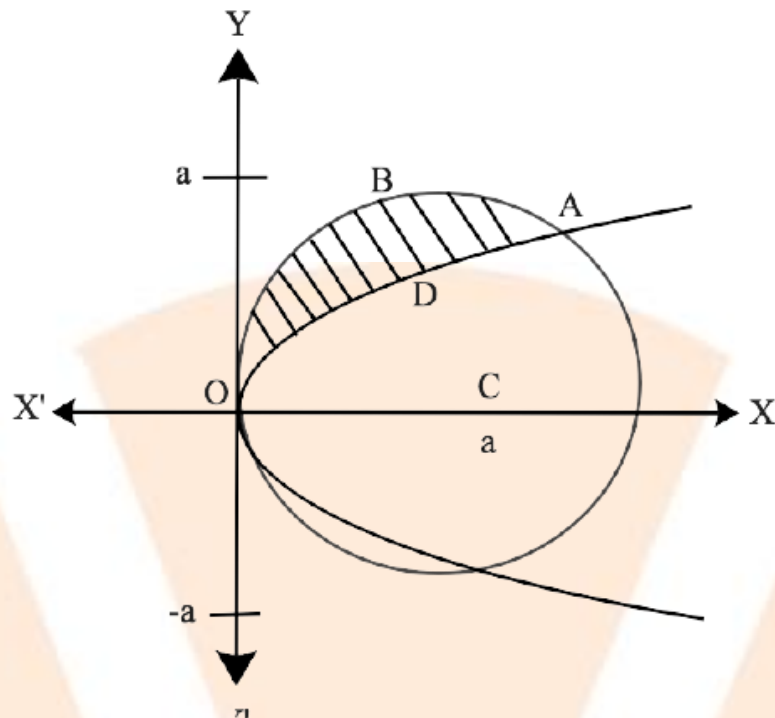
Required area = 2 area of ABCD

$$= 2 \int_a^{4a} y dx = 2 \times 2 \int_a^{4a} \sqrt{ax} dx$$

$$= 8\sqrt{a} \left| \frac{x^{3/2}}{3} \right|_a^{4a} = \frac{56}{3} a^2 \text{ sq. units}$$

8. Find the area of the region above the x-axis, included between the parabola $y^2 = ax$ and the circle $x^2 + y^2 = 2ax$.

Ans: Solving the given equations of curves, we have $x^2 + ax = 2ax$ or $x = 0$, $x = a$, which give $y = 0$, $y = \pm a$



From fig -

$$\text{Area ODAB} = \int_0^a (\sqrt{2ax - x^2} - \sqrt{ax}) dx$$

Let $x = 2a \sin^2 \theta$. Then $dx = 4a \sin \theta \cos \theta d\theta$ and $x = 0, \Rightarrow \theta = 0, x = a \Rightarrow \theta = \frac{\pi}{4}$.

$$\text{Again, } \int_0^a (\sqrt{2ax - x^2}) dx$$

$$= \int_0^{\frac{\pi}{4}} (2a \sin \theta \cos \theta)(4a \sin \theta \cos \theta) d\theta$$

$$= a^2 \int_0^{\frac{\pi}{4}} (1 - \cos 4\theta) d\theta = a^2 \left(\theta - \frac{\sin 4\theta}{4} \right)_0^{\frac{\pi}{4}} = \frac{\pi}{4} a^2$$

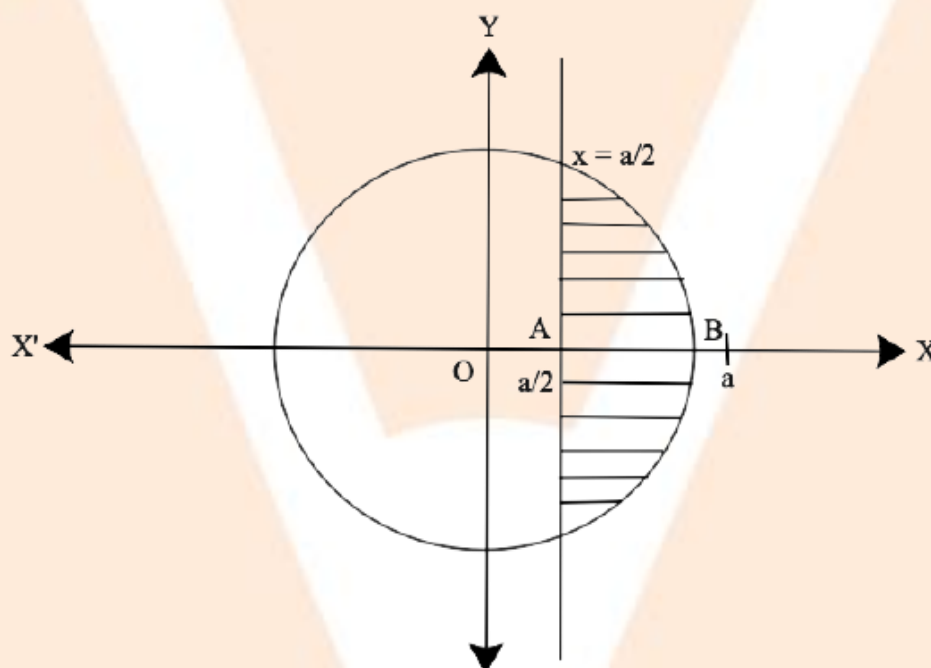
Furthermore,

$$\int_0^a (\sqrt{ax}) dx = \sqrt{a} \frac{2}{3} \left(x^{\frac{3}{2}} \right)_0^a = \frac{2}{3} a^2$$

$$\text{Thus the required area} = \frac{\pi}{4} a^2 - \frac{2}{3} a^2 = a^2 \left(\frac{\pi}{4} - \frac{2}{3} \right) \text{ sq units}$$

9. Find the area of a minor segment of the circle $x^2 + y^2 = a^2$ cut off by the line $x = \frac{a}{2}$.

Ans: Solving the equation $x^2 + y^2 = a^2$ and $x = \frac{a}{2}$, we obtain their points of intersection which are $\left(\frac{a}{2}, \sqrt{3}\frac{a}{2}\right)$ and $\left(\frac{a}{2}, -\sqrt{3}\frac{a}{2}\right)$.



Hence, from fig, we get -

$$\text{Required area} = 2 \times \text{area of OAB} = 2 \int_{a/2}^a \sqrt{a^2 - x^2} dx$$

$$= 2 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_{a/2}^a$$

$$= 2 \left[\frac{a^2}{2} \frac{\pi}{2} - \frac{a}{4} \cdot a \cdot \frac{\sqrt{3}}{2} - \frac{a^2}{2} \cdot \frac{\pi}{6} \right]$$

$$= \frac{a^2}{12} (6\pi - 3\sqrt{3} - 2\pi)$$

$$= \frac{a^2}{12} (4\pi - 3\sqrt{3}) \text{ sq. units}$$

Objective type questions

Choose the correct answer from the given four options in each of the Examples 10 to 12.

10. The area enclosed by the circle $x^2 + y^2 = 2$ is equal to

(A) 4π sq. units

(B) $2\sqrt{2}\pi$ sq. units

(C) $4\pi^2$ sq. units

(D) 2π sq. units

Ans: Correct answer is (D); since Area $= 4 \int_0^{\sqrt{2}} \sqrt{2-x^2} dx$

$$= 4 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^{\sqrt{2}} = 2\pi \text{ sq. units}$$

11. The area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is equal to

(A) $\pi^2 ab$

(B) πab

(C) $\pi a^2 b$

(D) πab^2

Ans: Correct answer is (B); since Area $= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$

$$= 4 \frac{b}{a} \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a = \pi ab \text{ sq. units}$$

12. The area of the region bounded by the curve $y = x^2$ and the line $y = 16$.

(A) $\frac{32}{3}$

(B) $\frac{256}{3}$

(C) $\frac{64}{3}$

(D) $\frac{128}{3}$

Ans: Correct answer is (B); since Area $= 2 \int_0^{16} \sqrt{y} dy$

$$= 2 \left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}} \right)_0^{16} = \frac{4}{3} \left(16^{\frac{3}{2}} - 0 \right)$$

$$= \frac{4}{3} \left(4^{2 \times \frac{3}{2}} - 0 \right)$$

$$= \frac{4^4}{3} = \frac{256}{3} \text{ sq. units}$$

Fill in the blanks in each of the Examples 13 and 14.

13. The area of the region bounded by the curve $x = y^2$, y-axis and the line $y=3$ and $y = 4$ is _____.

Ans: since Area $= \int_3^4 x dy = \int_3^4 y^2 dy$

$$= \left(\frac{y^3}{3} \right)_3^4 = \left(\frac{4^3}{3} - \frac{3^3}{3} \right)$$

$$= \left(\frac{64}{3} - \frac{27}{3} \right) = \frac{37}{3} \text{ sq. units}$$

14. The area of the region bounded by the curve $y = x^2 + x$, x-axis and the line $x = 2$ and $x = 5$ is equal to _____.

Ans: since Area $= \int_2^5 y dx = \int_2^5 (x^2 + x) dx$

$$= \left(\frac{x^3}{3} \right)_2^5 + \left(\frac{x^2}{2} \right)_2^5 = \left(\frac{5^3}{3} - \frac{2^3}{3} \right) + \left(\frac{5^2}{2} - \frac{2^2}{2} \right)$$

$$= \left(\frac{125}{3} - \frac{8}{3} \right) + \left(\frac{25}{2} - \frac{4}{2} \right)$$

$$= \left(\frac{117}{3} \right) + \left(\frac{21}{2} \right) = \frac{234+63}{6} = \frac{297}{6} \text{ sq. units}$$

EXERCISE

Short Answer Questions

1. Find the area of the region bounded by the curves $y^2 = 9x$, $y = 3x$.

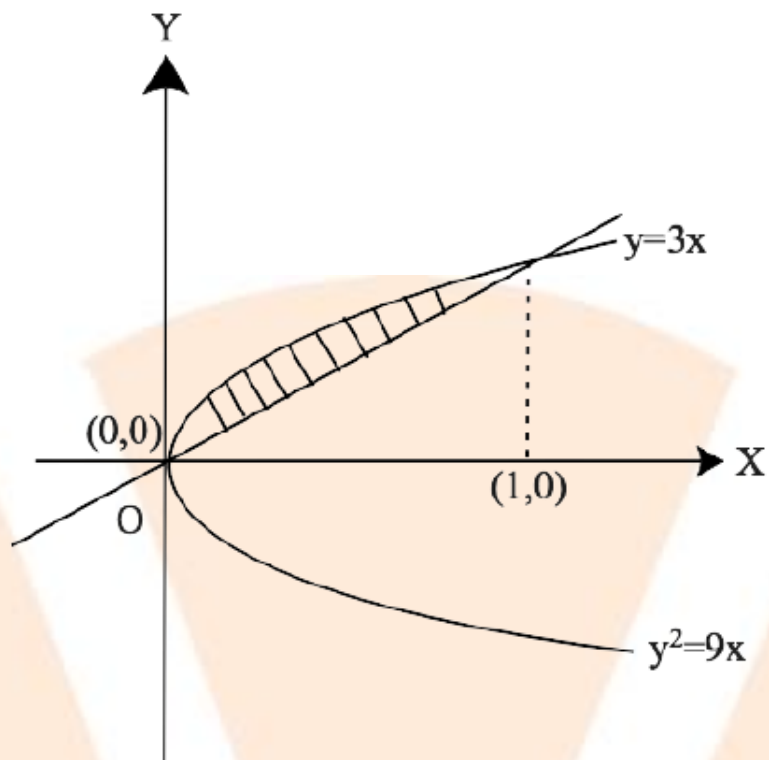
Ans: We have $y^2 = 9x$, $y = 3x$.

$$\Rightarrow (3x)^2 = 9x$$

$$\Rightarrow 9x^2 - 9x = 0$$

$$\Rightarrow 9x(x-1) = 0$$

$$\Rightarrow x = 0, 1$$



Therefore, required area = $\int_0^1 \sqrt{9x} dx - \int_0^1 3x dx$

$$\Rightarrow 3 \int_0^1 x^{\frac{1}{2}} dx - 3 \int_0^1 x dx$$

$$\Rightarrow 3 \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1 - 3 \left[\frac{x^2}{2} \right]_0^1$$

$$\Rightarrow 3 \left(\frac{2}{3} - 0 \right) - 3 \left(\frac{1}{2} - 0 \right)$$

$$\Rightarrow 2 - \frac{3}{2} = \frac{1}{2} \text{ units}$$

2. Find the area of the region bounded by the parabola $y^2 = 2px$, $x^2 = 2py$.

Ans: We have $y^2 = 2px$, $x^2 = 2py$.

$$y = \sqrt{2px}$$

$$\Rightarrow x^2 = 2p\sqrt{2px}$$

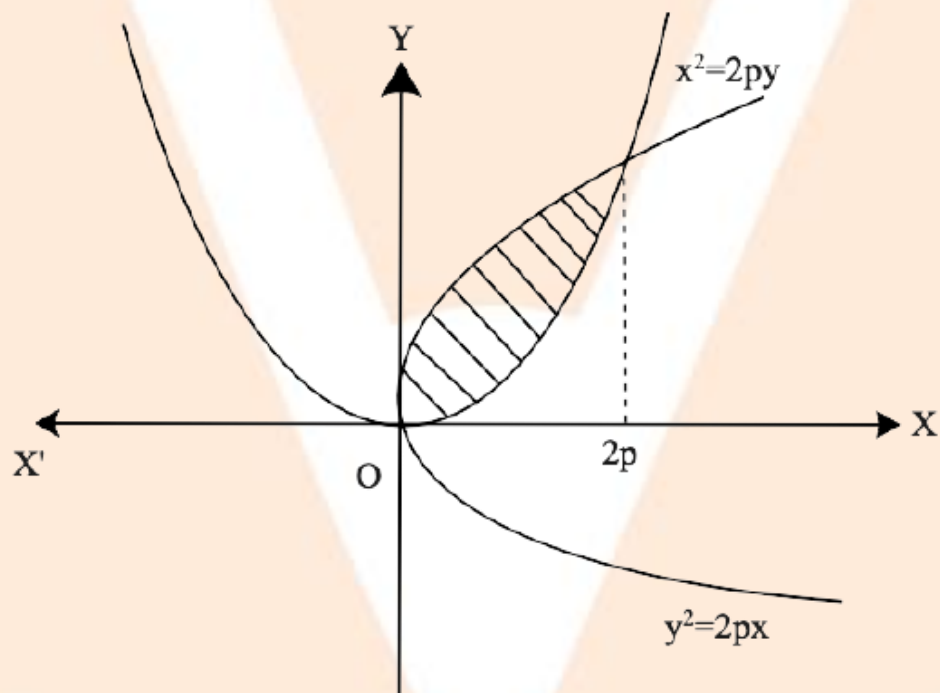
$$\Rightarrow x^4 = 4p^2(2px)$$

$$\Rightarrow x^4 = 8p^3x$$

$$\Rightarrow x^4 - 8p^3x = 0$$

$$\Rightarrow x(x^3 - 8p^3) = 0$$

$$\Rightarrow x = 0, 2p$$



$$\text{Required area} = \int_0^{2p} \sqrt{2px} \, dx - \int_0^{2p} \frac{x^2}{2p} \, dx$$

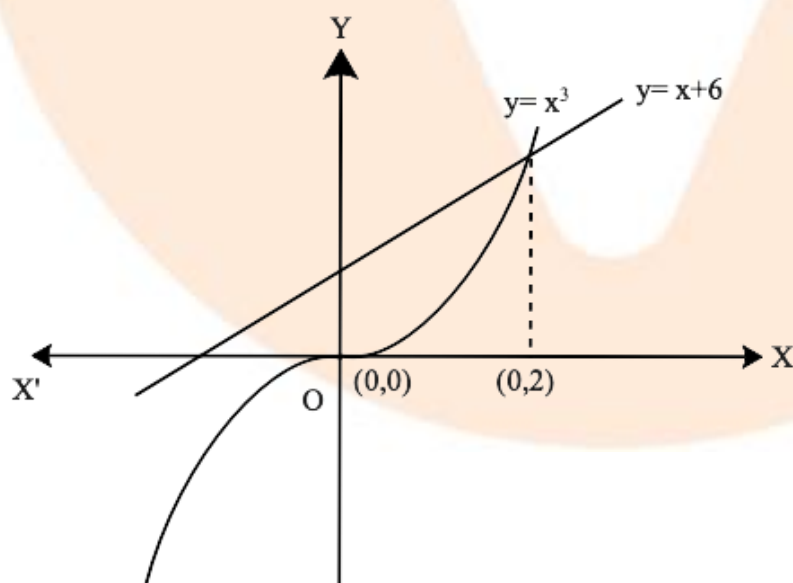
$$= \sqrt{2p} \int_0^{2p} \sqrt{x} \, dx - \frac{1}{2p} \int_0^{2p} x^2 \, dx$$

$$= \sqrt{2p} \left[\frac{2(x)^{\frac{3}{2}}}{3} \right]_0^{2p} - \frac{1}{2p} \left[\frac{(x)^3}{3} \right]_0^{2p}$$

$$\begin{aligned}
 &= \sqrt{2p} \left[\frac{2(2p)^{\frac{3}{2}}}{3} - 0 \right] - \frac{1}{2p} \left[\frac{(2p)^3}{3} - 0 \right] \\
 &= \sqrt{2p} \left(\frac{2}{3} \cdot 2\sqrt{2} p^{\frac{3}{2}} \right) - \frac{1}{2p} \left(\frac{1}{3} \cdot 8 p^3 \right) \\
 &= \frac{4\sqrt{2}}{3} \cdot \sqrt{2} p^2 - \frac{8}{6} p^2 \\
 &= \frac{8}{6} p^2 \\
 &= \frac{4}{6} p^2 \text{ sq unit}
 \end{aligned}$$

3. Find the area of the region bounded by the curve $y = x^3$ and $y = x + 6$ and $x = 0$.

Ans: We have, $y = x^3$ and $y = x + 6$ and $x = 0$.



Therefore, $x^3 = x + 6$

$$\Rightarrow x^3 - x = 6$$

$$\Rightarrow x^3 - x - 6 = 0$$

$$\Rightarrow x^2(x-2) + 2x(x-2) + 3(x-2) = 0$$

$$\Rightarrow (x-2)(x^2 + 2x + 3) = 0$$

$\Rightarrow x = 2$, With two imaginary points

Therefore, required area of shaded region = $\int_0^2 (x+6-x^3) dx$

$$= \left[\frac{x^2}{2} + 6x - \frac{x^4}{4} \right]_0^2$$

$$= \left[\frac{2^2}{2} + 6(2) - \frac{2^4}{4} - 0 \right]$$

$$= [2 + 12 - 4] = 10 \text{ sq units}$$

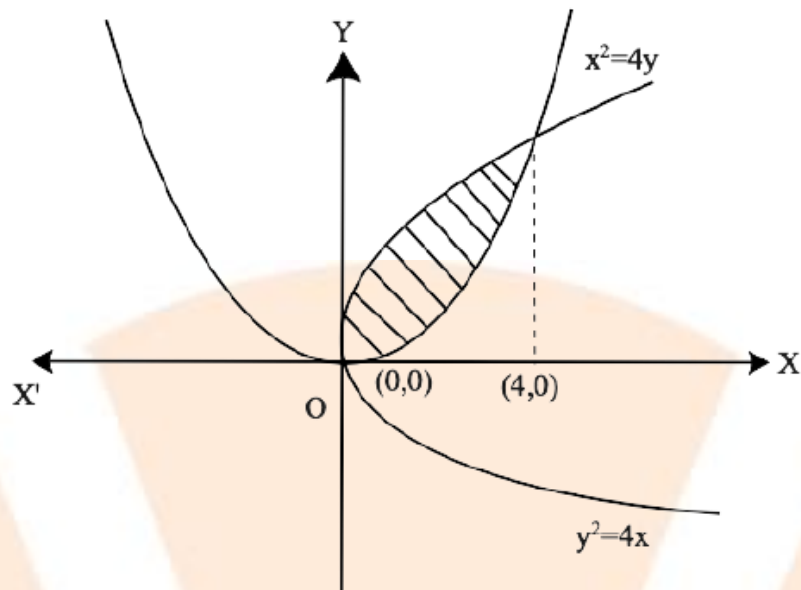
4. Find the area of the region bounded by the curve $y^2 = 4x$, $x^2 = 4y$.

Ans: Given equations of curves are

$$y^2 = 4x \text{ ----- (i)}$$

$$\text{and } x^2 = 4y \text{ ----- (ii)}$$

$$\Rightarrow \frac{x^2}{4} = y$$



$$\Rightarrow \frac{x^2}{4 \cdot 4} = 4x$$

$$\Rightarrow x^4 = 64x$$

$$\Rightarrow x^4 - 64x = 0$$

$$\Rightarrow x(x^3 - 64) = 0$$

$$\Rightarrow x(x^3 - 4^3) = 0$$

$$\Rightarrow x = 4, 0$$

Therefore, area of shaded region A = $\int_0^4 (\sqrt{4x} - \frac{x^2}{4}) dx$

$$= \int_0^4 (2\sqrt{x} - \frac{x^2}{4}) dx$$

$$= \left[\frac{2x^{\frac{3}{2}} \cdot 2}{3} - \frac{1}{4} \cdot \frac{x^3}{3} \right]_0^4$$

$$= \frac{32}{3} - \frac{16}{3} = \frac{16}{3} \text{ sq units}$$

5. Find the area of the region included between $y^2 = 9x$ and $y = x$

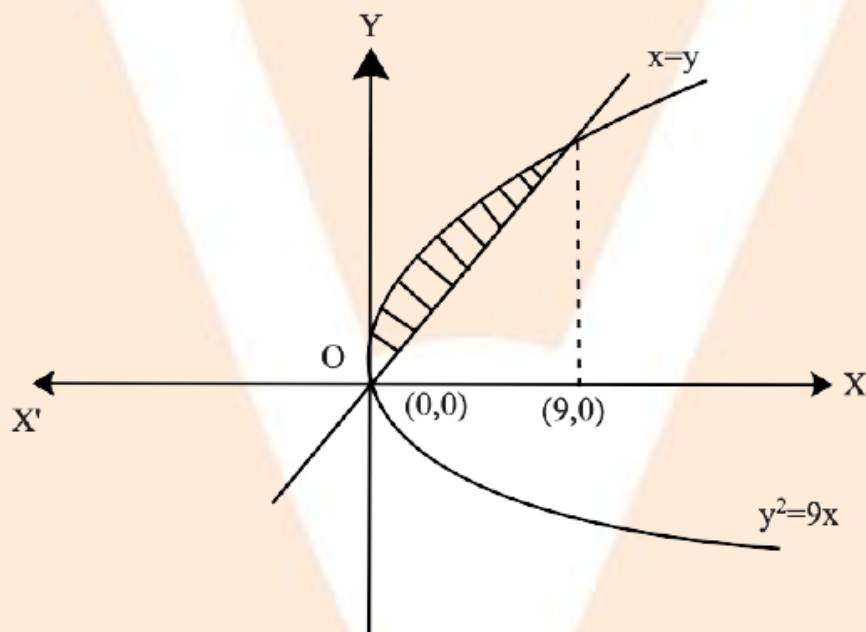
Ans: We have , $y^2 = 9x$ and $y = x$

$$\Rightarrow x^2 = 9x$$

$$\Rightarrow x^2 - 9x = 0$$

$$\Rightarrow x(x - 9) = 0$$

$$\Rightarrow x = 0, 9$$



Therefore, area of shaded region $A = \int_0^9 (\sqrt{9x} - x) dx$

$$= \int_0^9 (3\sqrt{x} - x) dx$$

$$= \int_0^9 (3\sqrt{x}) dx - \int_0^9 (x) dx$$

$$= \left[\frac{6x^{\frac{3}{2}}}{3} - \frac{x^2}{2} \right]_0^9$$

$$= \left(6 \cdot \frac{9^{\frac{3}{2}}}{3} - \frac{81}{2} - 0 \right)$$

$$= 54 - \frac{81}{2}$$

$$= \frac{108 - 81}{2}$$

$$= \frac{27}{2} \text{ sq units}$$

6. Find the area of the region enclosed by the parabola $x^2 = y$ and the line $y = x + 2$.

Ans: We have, $x^2 = y$ and $y = x + 2$

$$\Rightarrow x^2 = x + 2$$

$$\Rightarrow x^2 - (x + 2) = 0$$

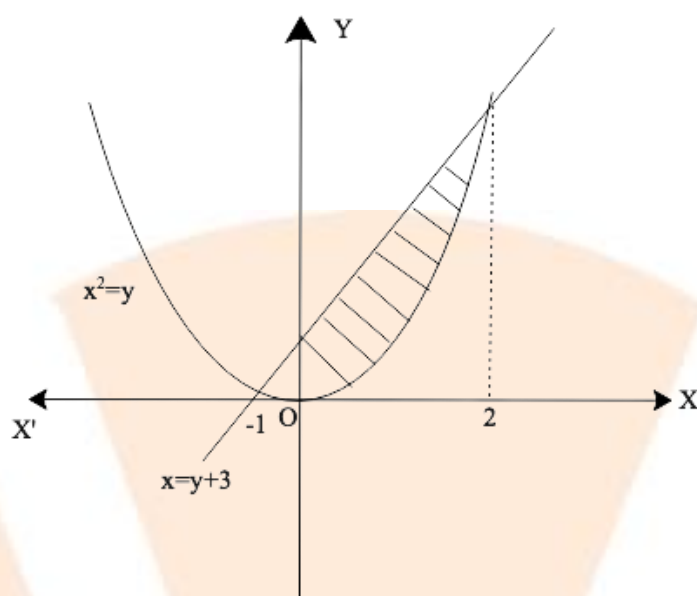
$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow x^2 - 2x + x - 2 = 0$$

$$\Rightarrow x(x - 2) + 1(x - 2) = 0$$

$$\Rightarrow (x + 1)(x - 2) = 0$$

$$\Rightarrow x = -1, 2$$



Therefore, area of shaded region $A = \int_{-1}^2 (x + 2 - x^2) dx$

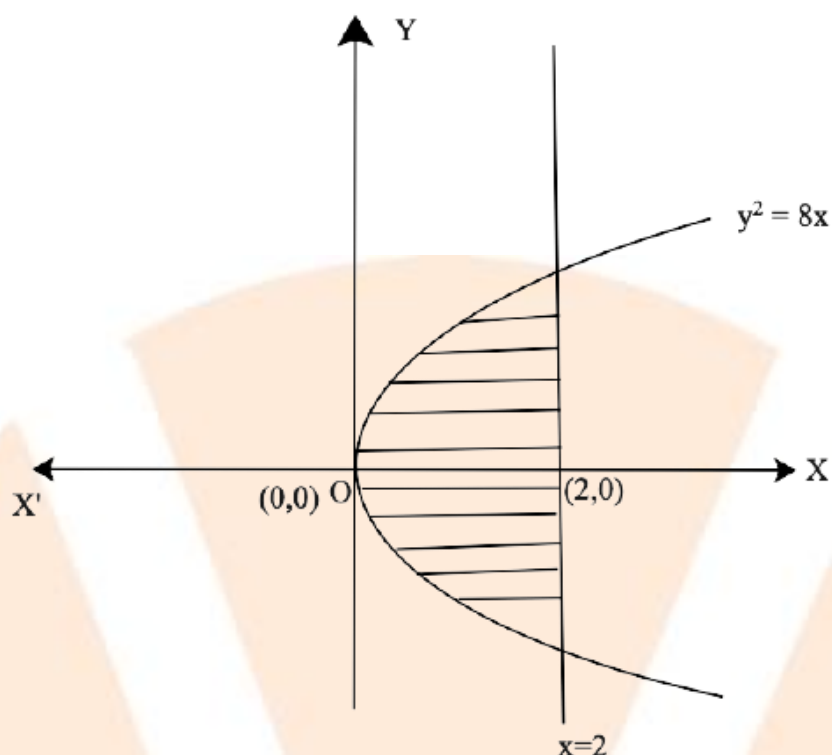
$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2$$

$$= \left[\frac{4}{2} + 4 - \frac{8}{3} - \frac{1}{2} + 2 - \frac{1}{3} \right]$$

$$= \left(8 - 3 - \frac{1}{2} \right) = \frac{9}{2} \text{ sq units}$$

7. Find the area of region bounded by the line $x = 2$ and the parabola $y^2 = 8x$.

Ans: We have, $y^2 = 8x$ and $x = 2$.



Therefore, area of shaded region $A = 2 \int_0^2 (\sqrt{8x}) dx$

$$= 2 \cdot 2\sqrt{2} \int_0^2 (\sqrt{x}) dx$$

$$= 4 \cdot \sqrt{2} \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^2$$

$$= 4 \cdot \sqrt{2} \left[\frac{2 \cdot 2^{\frac{3}{2}}}{3} \right]$$

$$= \frac{32}{3} \text{ sq units}$$

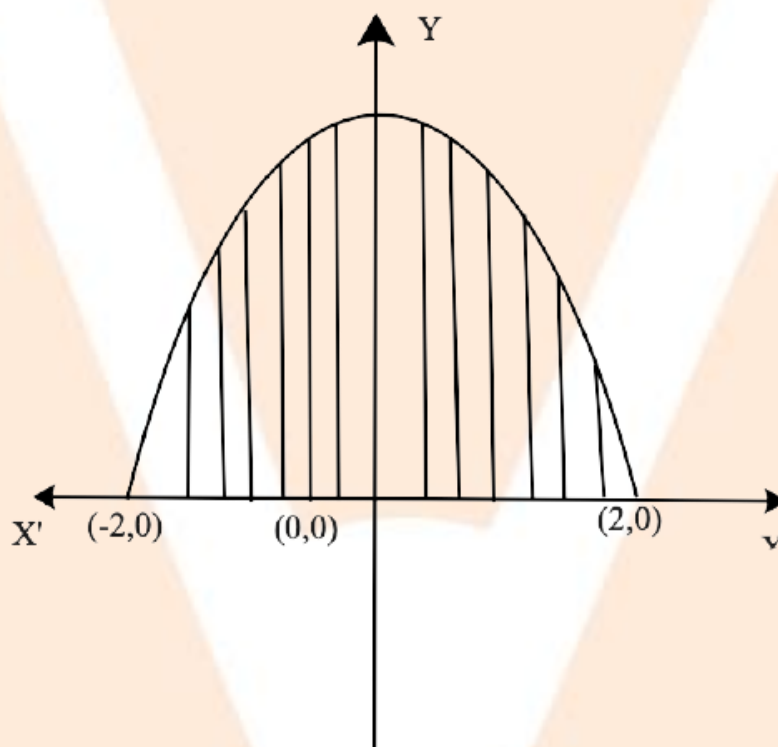
8. Sketch the region $\{(x, 0) : y = \sqrt{4-x^2}\}$ and x-axis. Find the area of the region using integration.

Ans: Given region is $\{(x, 0) : y = \sqrt{4-x^2}\}$ and x-axis

$$\text{We have, } y = \sqrt{4-x^2}$$

$$\Rightarrow y^2 = 4 - x^2$$

$$\Rightarrow y^2 + x^2 = 4$$



Therefore, area of shaded region $A = \int_{-2}^2 (\sqrt{4-x^2}) dx$

$$= \int_{-2}^2 (\sqrt{2^2 - x^2}) dx$$

$$= \left[\frac{x}{2} \sqrt{2^2 - x^2} + \frac{2^2}{2} \sin^{-1} \frac{x}{2} \right]_{-2}^2$$

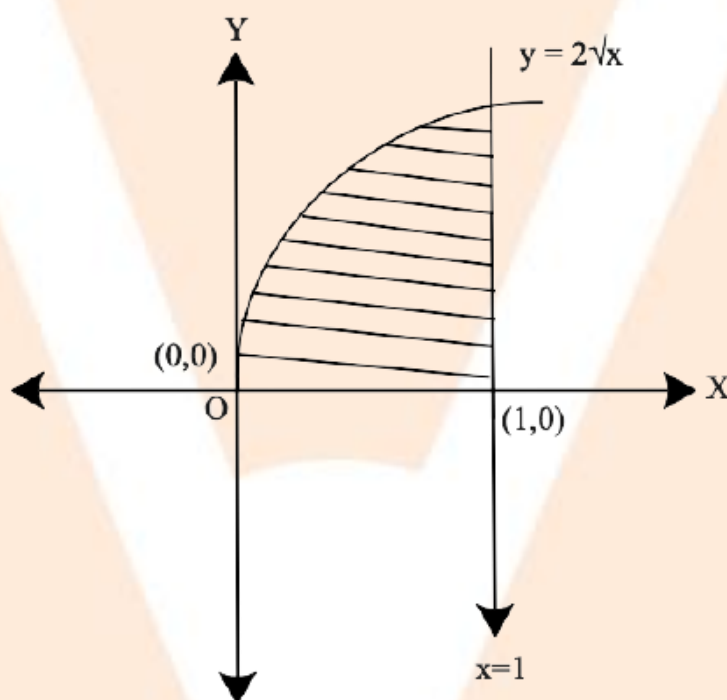
$$= \frac{2}{2} \cdot 0 + 2 \cdot \frac{\pi}{2} + \frac{2}{2} \cdot 0 - 2 \sin^{-1}(-1)$$

$$= 2 \cdot \frac{\pi}{2} + 2 \cdot \frac{\pi}{2}$$

$$= 2 \pi \text{ sq units}$$

9. Calculate the area under the curve $y = 2\sqrt{x}$ included between the lines $x = 0$ and $x = 1$.

Ans: We have, $y = 2\sqrt{x}$, $x = 0$ and $x = 1$.



Therefore, area of shaded region $A = \int_0^1 (2\sqrt{x}) dx$

$$= 2 \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1$$

$$= 2 \left(\frac{2}{3} \cdot 1 - 0 \right)$$

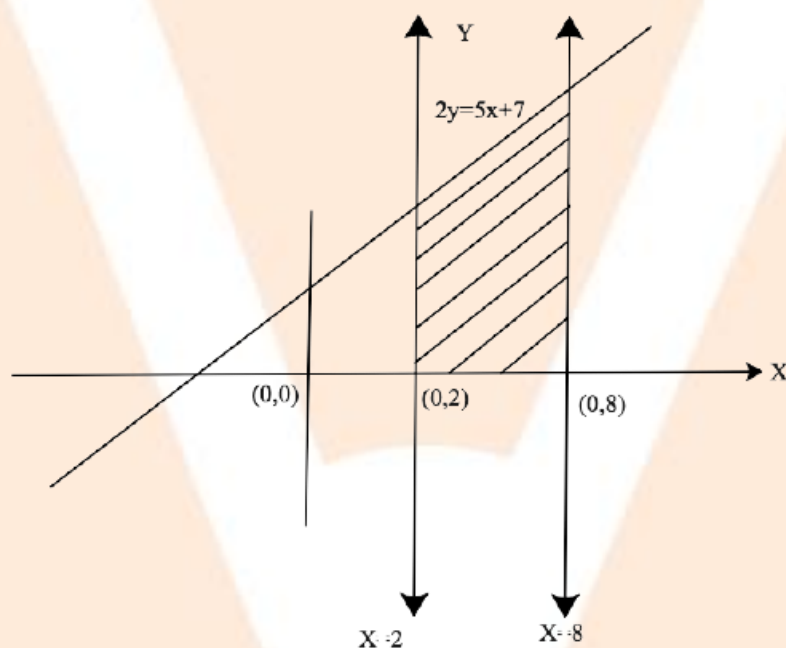
$$= \frac{4}{3} \text{ sq units}$$

10. Using integration, find the area of the region bounded by the line

$2y = 5x + 7$, x- axis and the lines $x = 2$ and $x = 8$.

Ans: We have $2y = 5x + 7$

$$\Rightarrow y = \frac{5x}{2} + \frac{7}{2}$$



$$\text{Area of shaded region} = \frac{1}{2} \int_2^8 (5x+7) dx$$

$$= \frac{1}{2} \left[\frac{5x^2}{2} + 7x \right]_2^8$$

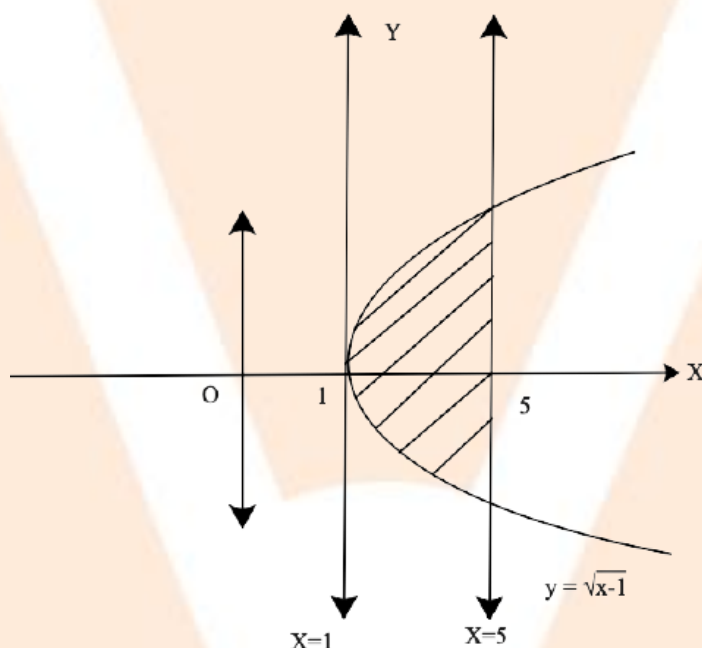
$$= \frac{1}{2} \left[\frac{5(8)^2}{2} + 7(8) - \frac{5(2)^2}{2} - 7(2) \right]$$

$$= 96 \text{ sq units}$$

11. Draw a rough sketch of the curve $y = \sqrt{x-1}$ in the interval $[1, 5]$. Find the area under the curve and between the lines $x = 1$ and $x = 5$.

Ans: Given equation of the curve is $y = \sqrt{x-1}$

$$\Rightarrow y^2 = x - 1$$



Therefore, area of shaded region $A = \int_1^5 (\sqrt{x-1}) dx$

$$= \left[\frac{2(x-1)^{\frac{3}{2}}}{3} \right]_1^5$$

$$= \left(\frac{2}{3} \cdot (5-1)^{\frac{3}{2}} - 0 \right)$$

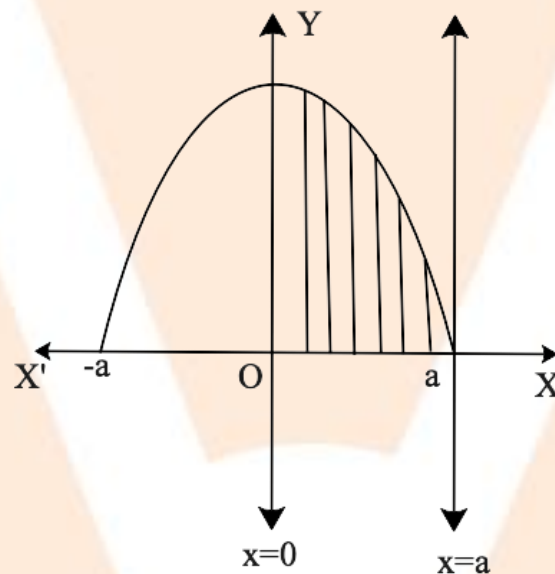
$$= \frac{16}{3} \text{ sq units}$$

12. Determine the area under the curve $y = \sqrt{a^2 - x^2}$ included between the lines $x = 0$ and $x = a$.

Ans: Given region is $y = \sqrt{a^2 - x^2}$

$$\Rightarrow y^2 = a^2 - x^2$$

$$\Rightarrow y^2 + x^2 = a^2$$



Therefore, area of shaded region $A = \int_0^a (\sqrt{a^2 - x^2}) dx$

$$= \int_0^a (\sqrt{a^2 - x^2}) dx$$

$$= \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

$$= \left[0 + \frac{a^2}{2} \sin^{-1} \frac{a}{a} - 0 - \frac{a^2}{2} \sin^{-1} 0 \right]$$

$$= \frac{a^2}{2} \cdot \frac{\pi}{2}$$

$$= \frac{a^2}{4} \pi \text{ sq units}$$

13. Find the area of the region bounded by $y = \sqrt{x}$ and $y = x$.

Ans: Given equations are $y = \sqrt{x}$ and $y = x$.

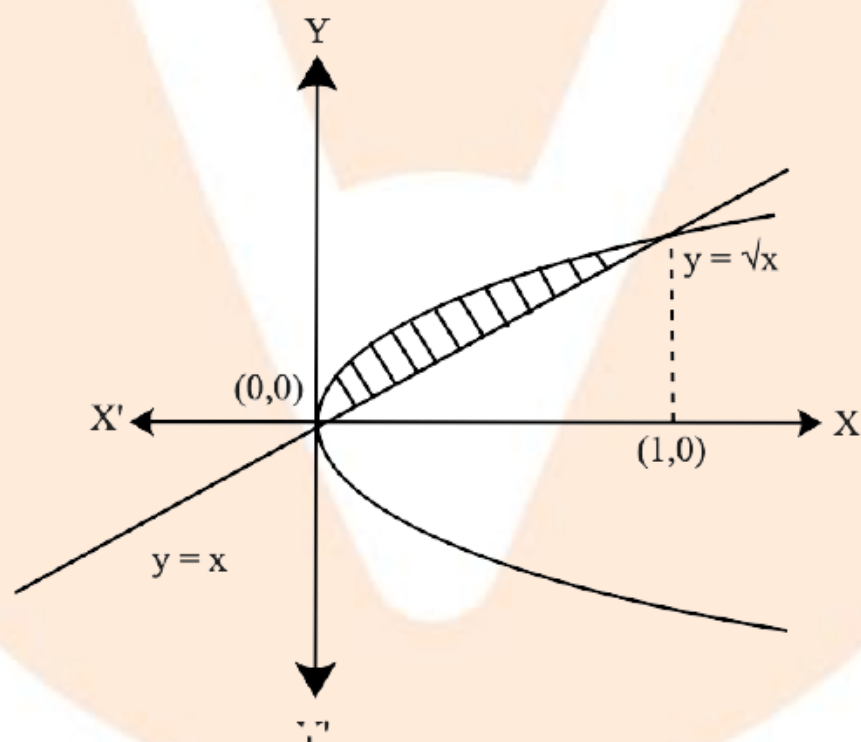
$$\Rightarrow x = \sqrt{x}$$

$$\Rightarrow x^2 = x$$

$$\Rightarrow x^2 - x = 0$$

$$\Rightarrow x(x - 1) = 0$$

$$\Rightarrow x = 0, 1$$



Therefore, area of shaded region $A = \int_0^1 (\sqrt{x}) dx - \int_0^1 (x) dx$

$$= \left[\frac{2(x)^{\frac{3}{2}}}{3} \right]_0^1 - \left[\frac{(x)^2}{2} \right]_0^1$$

$$= \left(\frac{2}{3} \cdot 1 - \frac{1}{2} \right)$$

$$= \frac{2}{3} - \frac{1}{2}$$

$$= \frac{1}{6} \text{ sq units}$$

14. Find the area enclosed by the curve $y = -x^2$ and the straight line $x + y + 2 = 0$

Ans: We have $y = -x^2$ and the straight line $x + y + 2 = 0$.

$$\Rightarrow -x - 2 = -x^2$$

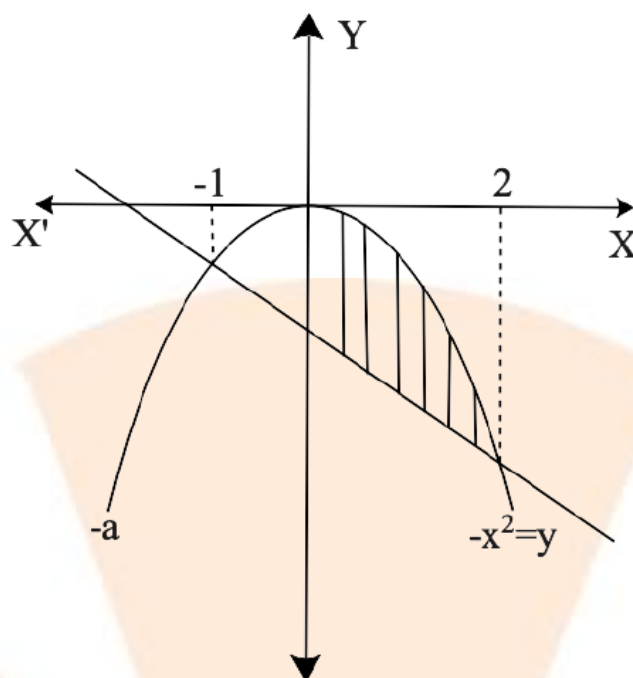
$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow x^2 + x - 2x - 2 = 0$$

$$\Rightarrow x(x+1) - 2(x+1) = 0$$

$$\Rightarrow (x-2)(x+1) = 0$$

$$\Rightarrow x = 2, -1.$$



$$\text{Area of the shaded region} = \left| \int_{-1}^2 (-x-2+x^2) dx \right|$$

$$= \left| \int_{-1}^2 (x^2 - x - 2) dx \right|$$

$$= \left[\left[\frac{(x)^3}{3} - \frac{(x)^2}{2} - 2x \right] \right]_{-1}^2$$

$$= \left[\left[\frac{8}{3} - \frac{4}{2} - 4 + \frac{1}{3} + \frac{1}{2} - 2 \right] \right]$$

$$= \left| \frac{16-12-24+2+3-12}{6} \right| = \left| \frac{-27}{6} \right|$$

$$= \frac{9}{2} \text{ sq units}$$

15. Find the area bounded by the curve $y = \sqrt{x}$, $x = 2y + 3$ in the first quadrant and x- axis.

Ans: Given equation of the curves for $y = \sqrt{x}$, $x = 2y + 3$ in the first quadrant.

On solving both the equations for y , we get

$$y = \sqrt{2y+3}$$

$$\Rightarrow y^2 = 2y+3$$

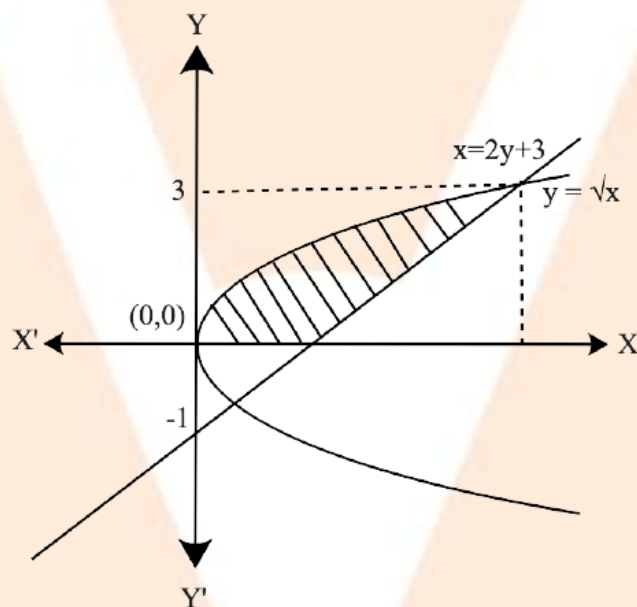
$$\Rightarrow y^2 - 2y - 3 = 0$$

$$\Rightarrow y^2 - 3y + y - 3 = 0$$

$$\Rightarrow y(y-3) + 1(y-3) = 0$$

$$\Rightarrow (y+1)(y-3) = 0$$

$$\Rightarrow y = -1, 3$$



$$\text{Area of shaded region} = \int_0^3 (2y+3-y^2) dy$$

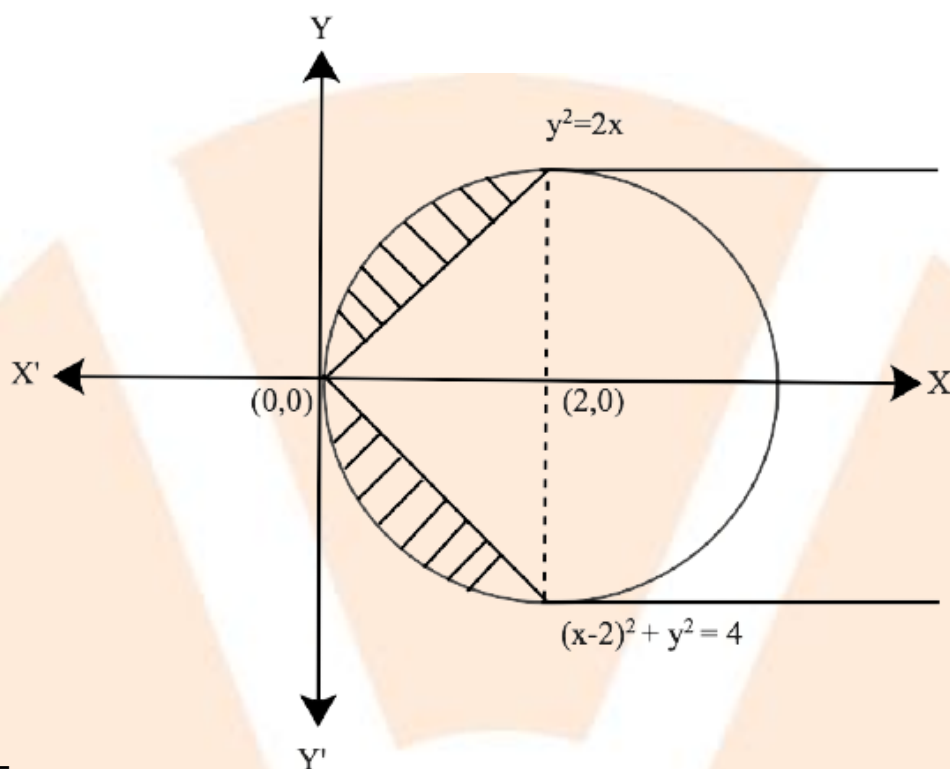
$$= \left[\frac{2y^2}{2} + 3y - \frac{y^3}{3} \right]_0^3$$

$$= \left[\frac{18}{2} + (9) - 9 - 0 \right]$$

$$= 9 \text{ sq units}$$

16. Find the area of the region bounded by the curve $y^2 = 2x$ and $x^2 + y^2 = 4x$.

Ans: We have $y^2 = 2x$ and $x^2 + y^2 = 4x$



$$\Rightarrow x^2 + 2x = 4x$$

$$\Rightarrow x^2 - 2x = 0$$

$$\Rightarrow x(x - 2) = 0$$

$$\Rightarrow x = 0 \text{ Or } x = 2$$

$$\text{Also, } x^2 + y^2 = 4x$$

$$\Rightarrow x^2 - 4x = -y^2$$

$$\Rightarrow x^2 - 4x + 4 = -y^2 + 4$$

$$\Rightarrow (x - 2)^2 - 2^2 = -y^2$$

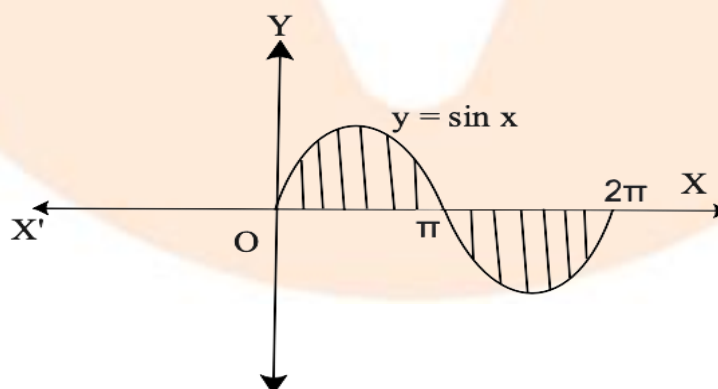
$$\therefore \text{ Required area} = 2 \int_0^2 \left[\sqrt{2^2 - (x - 2)^2} - \sqrt{2x} \right] dx$$

$$\begin{aligned}
 &= 2 \left[\left\{ \frac{x-2}{2} \cdot \sqrt{2^2 - (x-2)^2} + \frac{2^2}{2} \sin^{-1} \left(\frac{x-2}{2} \right) \right\}_0^2 - \left\{ \sqrt{2} \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right\}_0^2 \right] \\
 &= 2 \left[\left(0 + 0 - 1 \cdot 0 + 2 \cdot \frac{\pi}{2} \right) - \frac{2\sqrt{2}}{3} \left(2^{\frac{3}{2}} - 0 \right) \right] \\
 &= \frac{4\pi}{2} - \frac{8 \cdot 2}{3} \\
 &= 2\pi - \frac{16}{3} \\
 &= 2 \left(\pi - \frac{8}{3} \right) \text{sq. units}
 \end{aligned}$$

17. Find the area bounded by the curve $y = \sin x$ between $x=0$ and $x=2\pi$.

Ans: Area of the shaded region = $\int_0^{2\pi} (\sin x) dx = \int_0^{\pi} (\sin x) dx + \left| \int_{\pi}^{2\pi} (\sin x) dx \right|$

$$\begin{aligned}
 &= [-\cos x]_0^{\pi} + \left| [-\cos x]_{\pi}^{2\pi} \right| \\
 &= -[\cos \pi - \cos 0] + \left| -[\cos 2\pi - \cos \pi] \right|
 \end{aligned}$$

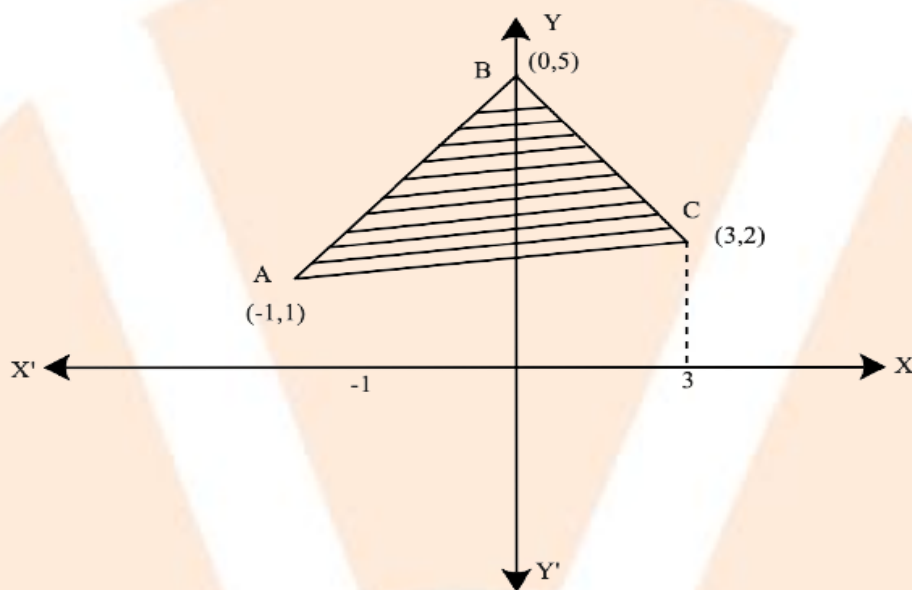


$$= -[-1-1] + \left| -(1+1) \right|$$

$$= 2+2 = 4 \text{ sq units}$$

18. Find the area of the region bounded by the triangle whose vertices are $(-1, 1)$, $(0, 5)$ and $(3, 2)$, using integration.

Ans: Let we have the vertices of a $\triangle ABC$ as $A(-1, 1)$, $B(0, 5)$ and $C(3, 2)$.



Equation of AB is $y - 1 = \left(\frac{5-1}{0-1} \right) (x + 1)$

$$\Rightarrow y - 1 = 4x + 4$$

$$\Rightarrow y = 4x + 5 \text{ --- (i)}$$

And equation of BC is $y - 5 = \left(\frac{2-5}{3-0} \right) (x - 0)$

$$\Rightarrow y - 5 = -\frac{3}{3} (x)$$

$$\Rightarrow y = 5 - x \text{ --- (ii)}$$

Similarly, equation of AC is $y - 1 = \left(\frac{2-1}{3+1} \right) (x + 1)$

$$\Rightarrow y-1=\frac{1}{4}(x+1)$$

$$\Rightarrow 4y = x + 5 \text{ --- (iii)}$$

$$\text{Area of shaded region} = \int_{-1}^0 (y_1 - y_2) dx + \int_0^3 (y_1 - y_2) dx$$

$$= \int_{-1}^0 \left(4x + 5 - \frac{x+5}{4} \right) dx + \int_0^3 \left(5 - x - \frac{x+5}{4} \right) dx$$

$$= \left[\frac{4x^2}{2} + 5x - \frac{x^2}{8} - \frac{5x}{4} \right]_{-1}^0 + \left[5x - \frac{x^2}{2} - \frac{x^2}{8} - \frac{5x}{4} \right]_0^3$$

$$= \left[0 - \left(\frac{4(-1)^2}{2} + 5(-1) - \frac{(-1)^2}{8} - \frac{5(-1)}{4} \right) \right] + \left[5(3) - \frac{3^2}{2} - \frac{3^2}{8} - \frac{5(3)}{4} - 0 \right]$$

$$= \left[0 - \left(2 - 5 - \frac{1}{8} + \frac{5}{4} \right) \right] + \left[15 - \frac{9}{2} - \frac{9}{8} - \frac{15}{4} \right]$$

$$= 3 + \frac{1}{8} - \frac{5}{4} + 15 - \frac{9}{2} - \frac{9}{8} - \frac{15}{4}$$

$$= 18 + \left(\frac{1 - 10 - 36 - 9 - 30}{8} \right)$$

$$= 18 - \frac{84}{8}$$

$$= 18 - \frac{21}{2}$$

$$= \frac{15}{2} \text{ sq. units}$$

19. Draw a rough sketch of the region $\{(x, y) : y^2 \leq 6ax \text{ and } x^2 + y^2 \leq 16a^2\}$. Also find the area of the region sketched using method of integration.

Ans: We have, $y^2 = 6ax$ and $x^2 + y^2 = 16a^2$

$$\Rightarrow x^2 + 6ax = 16a^2$$

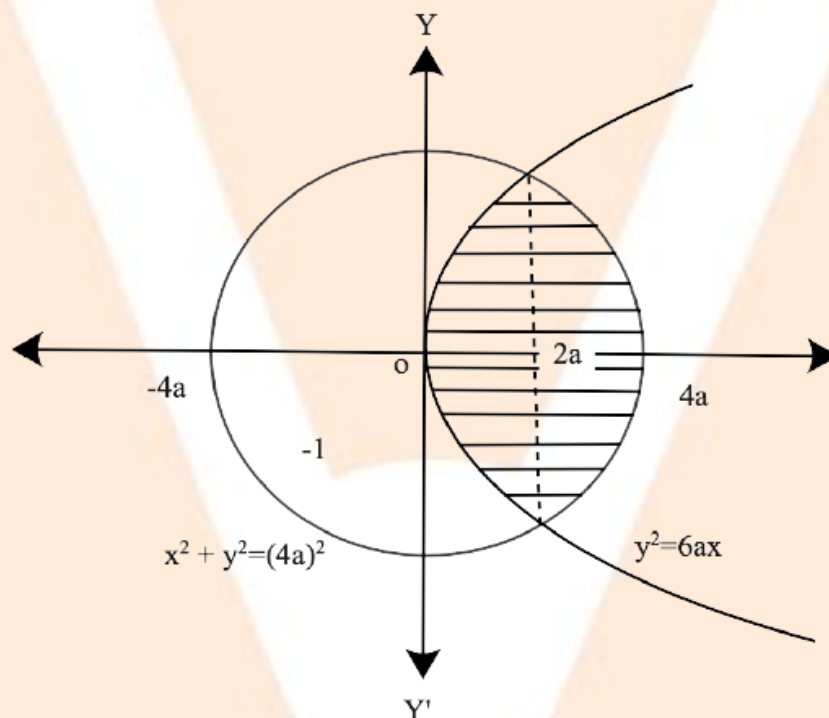
$$\Rightarrow x^2 + 6ax - 16a^2 = 0$$

$$\Rightarrow x^2 + 8ax - 2ax - 16a^2 = 0$$

$$\Rightarrow x(x + 8a) - 2a(x + 8a) = 0$$

$$\Rightarrow (x - 2a)(x + 8a) = 0$$

$$\Rightarrow x = 2a, -8a$$



$$\text{Area of required region} = 2 \left[\int_0^{2a} \sqrt{6ax} dx + \int_{2a}^{4a} \sqrt{(4a)^2 - x^2} dx \right]$$

$$= 2 \left[\sqrt{6a} \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^{2a} + \left[\frac{x}{2} \sqrt{(4a)^2 - x^2} + \frac{(4a)^2}{2} \sin^{-1} \frac{x}{4a} \right]_{2a}^{4a} \right]$$

$$= 2 \left[\sqrt{6a} \times \frac{2}{3} (2a)^{\frac{3}{2}} + \frac{4a}{2} (0) + \frac{16a^2}{2} \times \frac{\pi}{2} - \frac{2a}{2} \sqrt{16a^2 - 4a^2} - \frac{16a^2}{2} \sin^{-1} \frac{2a}{4a} \right]$$

$$= 2 \left[\sqrt{6a} \times \frac{2}{3} \times 2\sqrt{2}a^{\frac{3}{2}} + 0 + 4\pi a^2 - a \cdot 2\sqrt{3}a - 8a^2 \cdot \frac{\pi}{6} \right]$$

$$\begin{aligned}
 &= 2 \left[\sqrt{12} \times \frac{4}{3} a^2 + 4\pi a^2 - 2\sqrt{3}a^2 - \frac{4a^2\pi}{3} \right] \\
 &= 2 \left[\frac{8\sqrt{3}a^2 + 12\pi a^2 - 6\sqrt{3}a^2 - 4a^2\pi}{3} \right] \\
 &= \frac{2}{3} a^2 [8\sqrt{3} + 12\pi - 6\sqrt{3} - 4\pi] \\
 &= \frac{2}{3} a^2 [2\sqrt{3} + 8\pi] \\
 &= \frac{4}{3} a^2 [\sqrt{3} + 4\pi] \text{ sq. units}
 \end{aligned}$$

20. Compute the area bounded by the lines $x + 2y = 2$, $y - x = 1$ and $2x + y = 7$.

Ans: We have,

$$x + 2y = 2 \text{ -----(i)}$$

$$y - x = 1 \text{ -----(ii)}$$

$$\text{and } 2x + y = 7 \text{ -----(iii)}$$

On solving Eqs. (i) and (ii), we get

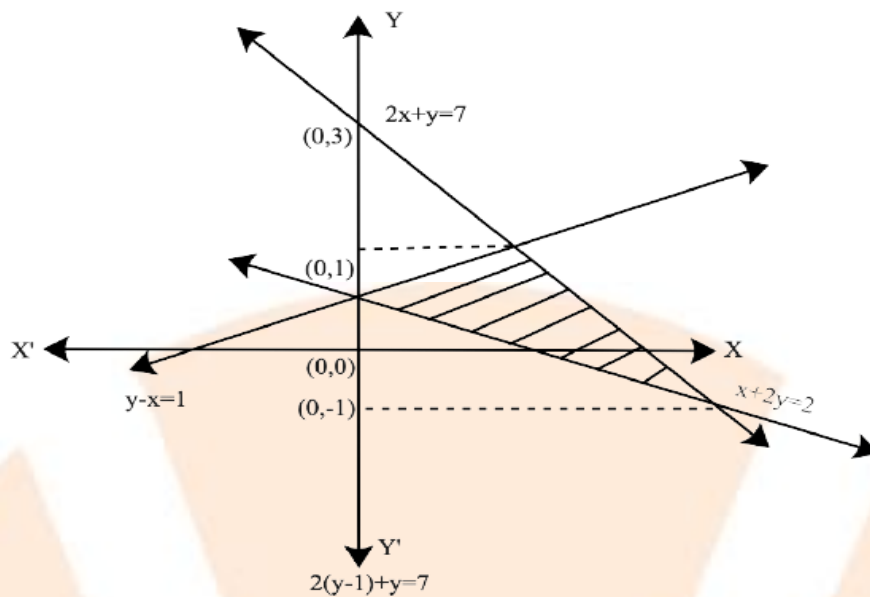
$$y - (2 - 2y) = 1$$

$$\Rightarrow 3y - 2 = 1$$

$$\Rightarrow y = 1$$

From eq. (i)-

$$x = 2 - 2(1) = 0$$



On solving Eqs. (ii) and (iii), we get

$$2(y-1) + y = 7$$

$$\Rightarrow 2y - 2 + y = 7$$

$$\Rightarrow 3y = 9$$

$$\Rightarrow y = 3$$

From eq. (ii)-

$$x = 3 - 1 = 2$$

On solving Eqs. (i) and (iii), we get

$$2(2 - 2y) + y = 7$$

$$\Rightarrow 4 - 4y + y = 7$$

$$\Rightarrow -3y = 3$$

$$\Rightarrow y = -1$$

From eq. (iii)-

$$2x = 7 - (-1)$$

$$\Rightarrow x = 4$$

$$\begin{aligned}
 \text{Required area} &= \int_{-1}^3 \frac{7-y}{2} dy - \left[\int_{-1}^1 (2-2y) dy + \int_1^3 (y-1) dy \right] \\
 &= \frac{1}{2} \left(7y - \frac{y^2}{2} \right)_{-1}^3 - \left[(2y - y^2)_{-1}^1 + \left(\frac{y^2}{2} - y \right)_{-1}^3 \right] \\
 &= \frac{1}{2} \left(21 - \frac{9}{2} + 7 + \frac{1}{2} \right) - (2 - 1 + 2 + 1) - \left(\frac{9}{2} - 3 - \frac{1}{2} + 1 \right) \\
 &= 12 - 4 - 2 = 6 \text{ sq. units}
 \end{aligned}$$

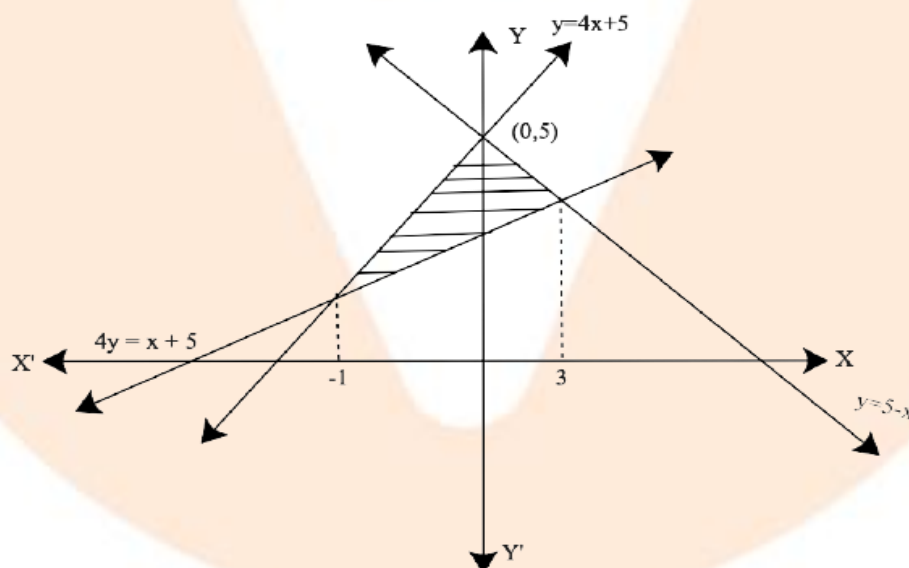
21. Find the area bounded by the lines $y = 4x + 5$, $y = 5 - x$ and $4y = x + 5$.

Ans: Given equations of lines are

$y = 4x + 5$ -----(i)

$y = 5 - x$ -----(ii)

and $4y = x + 5$ -----(iii)



On solving eq (i) and (ii), we get

$$4x + 5 = 5 - x$$

$$\Rightarrow 5x = 0$$

$$\Rightarrow x = 0$$

On solving eq (i) and (iii), we get

$$4(4x + 5) = 5 + x$$

$$\Rightarrow 16x + 20 = 5 + x$$

$$\Rightarrow 15x = -15$$

$$\Rightarrow x = -1$$

On solving eq (ii) and (iii), we get

$$4(5 - x) = 5 + x$$

$$\Rightarrow 20 - 4x = 5 + x$$

$$\Rightarrow 15 = 5x$$

$$\Rightarrow x = 3$$

$$\text{Required area} = \int_{-1}^0 (4x + 5) dx + \int_0^3 (5 - x) dx - \frac{1}{4} \int_{-1}^3 (x + 5) dx$$

$$= \left[\frac{4x^2}{2} + 5x \right]_{-1}^0 + \left[5x - \frac{x^2}{2} \right]_0^3 - \frac{1}{4} \left[\frac{x^2}{2} + 5x \right]_{-1}^3$$

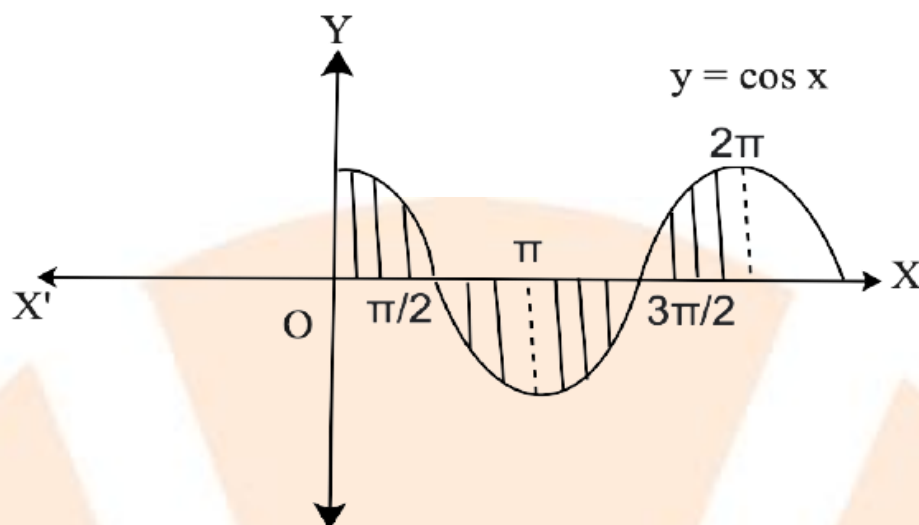
$$= [0 - 2 + 5] + \left[15 - \frac{9}{2} - 0 \right] - \frac{1}{4} \left[\frac{9}{2} + 15 - \frac{1}{2} + 5 \right]$$

$$= 3 + \frac{21}{2} - \frac{1}{4} \times 24$$

$$= -3 + \frac{21}{2} = \frac{15}{2} \text{ sq. units}$$

22. Find the area bounded by the curve $y = 2\cos x$ and the x-axis from $x = 0$ to $x = 2\pi$.

Ans:



Required area of shaded region $= \int_0^{2\pi} 2\cos x dx$

$$= \int_0^{\frac{\pi}{2}} 2\cos x dx + \left| \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 2\cos x dx \right| + \int_{\frac{3\pi}{2}}^{2\pi} 2\cos x dx$$

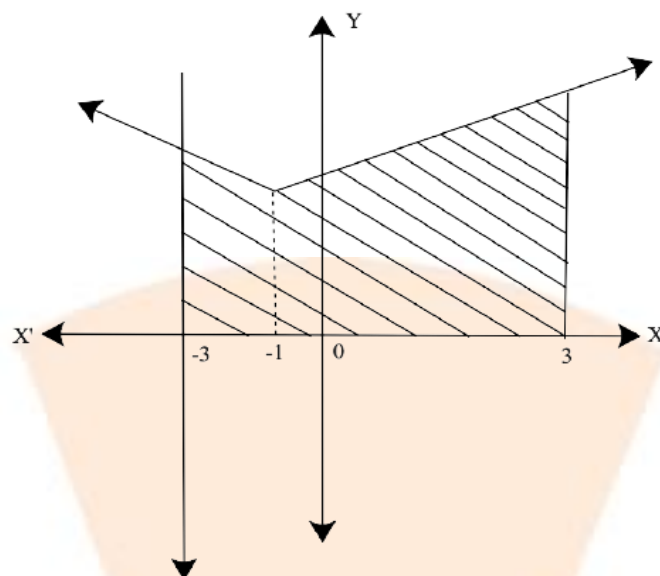
$$= 2 \left[\sin x \right]_0^{\frac{\pi}{2}} + \left| 2 \left[\sin x \right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \right| + 2 \left[\sin x \right]_{\frac{3\pi}{2}}^{2\pi}$$

$$= 2(1-0) + |2(-1-1)| + 2(0+1)$$

$$= 2 + 4 + 2 = 8 \text{ sq. units}$$

23. Draw a rough sketch of the given curve $y = 1 + |x + 1|$, $x = -3$, $x = 3$, $y = 0$ and find the area of the region bounded by them, using integration.

Ans: We have, $y = 1 + |x + 1|$, $x = -3$, $x = 3$, $y = 0$



$$y = \begin{cases} -x, & \text{if } x < -1 \\ x + 2, & \text{if } x \geq -1 \end{cases}$$

Area of shaded region, $A = \int_{-3}^{-1} -x dx + \int_{-1}^3 (x + 2) dx$

$$= -\left[\frac{x^2}{2}\right]_{-3}^{-1} + \left[\frac{x^2}{2} + 2x\right]_{-1}^3$$

$$= -\left[\frac{1}{2} - \frac{9}{2}\right] + \left[\frac{9}{2} + 6 - \frac{1}{2} + 2\right]$$

$$= -(-4) + (8 + 4)$$

$$= 4 + 12 = 16 \text{ sq. units}$$

Choose the correct answer from the given four options in each of the Exercises 24 to 34.

24. The area of the region bounded by the y- axis, $y = \cos x$ and $y = \sin x$,

$0 \leq x \leq \frac{\pi}{2}$ is

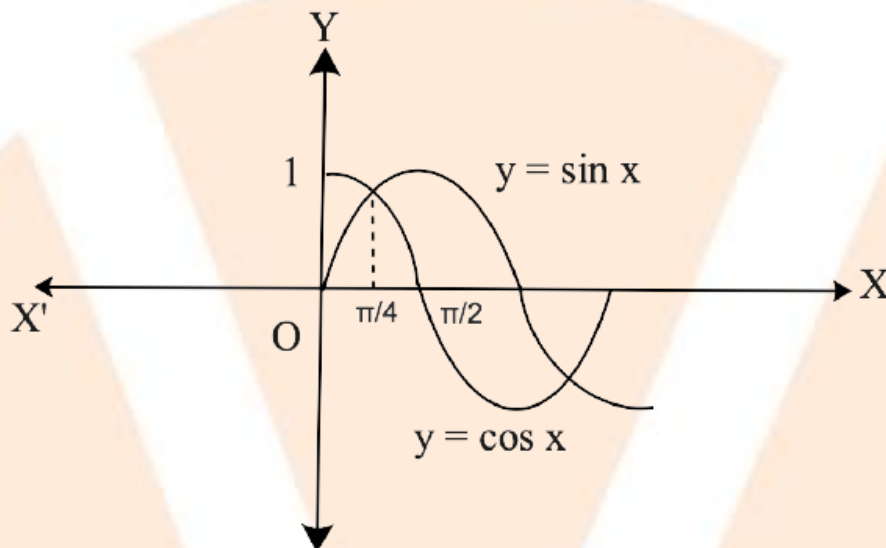
(A) $\sqrt{2}$ sq units

(B) $(\sqrt{2} + 1)$ sq units

(C) $(\sqrt{2} - 1)$ sq units

(D) $(2\sqrt{2} - 1)$ sq units

Ans: We have, y-axis i.e $x=0$, $y=\cos x$ and $y=\sin x$, where $0 \leq x \leq \frac{\pi}{2}$



Therefore, required area = $\int_0^{\pi/4} (\cos x - \sin x) dx$

$$= [\sin x]_0^{\pi/4} + [\cos x]_0^{\pi/4}$$

$$= \left(\sin \frac{\pi}{4} - \sin 0\right) + \left(\cos \frac{\pi}{4} - \cos 0\right)$$

$$= \left(\frac{1}{\sqrt{2}} - 0\right) + \left(\frac{1}{\sqrt{2}} - 1\right)$$

$$= \frac{2}{\sqrt{2}} - 1$$

$$= (\sqrt{2} - 1) \text{ sq units}$$

25. The area of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is

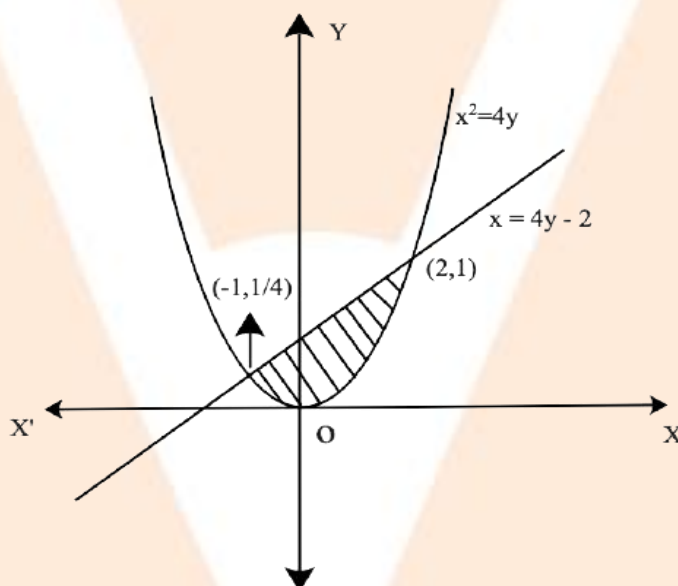
(A) $\frac{3}{8}$ sq units

(B) $\frac{5}{8}$ sq units

(C) $\frac{7}{8}$ sq units

(D) $\frac{9}{8}$ sq units

Ans: Given equation of curve is $x^2 = 4y$ and the straight line $x = 4y - 2$



For intersection point , put $x = 4y - 2$ in equation of curve , we get

$$(4y - 2)^2 = 4y$$

$$\Rightarrow 16y^2 + 4 - 16y = 4y$$

$$\Rightarrow 16y^2 - 20y + 4 = 0$$

$$\Rightarrow 4y^2 - 5y + 1 = 0$$

$$\Rightarrow 4y^2 - 4y - y + 1 = 0$$

$$\Rightarrow 4y(y-1) - 1(y-1) = 0$$

$$\Rightarrow (4y-1)(y-1) = 0$$

$$\text{Therefore, } y = 1, \frac{1}{4}$$

For $y = 1$, $x = \sqrt{4 \cdot 1} = 2$ [since, negative value does not satisfy the equation of line]

For $y = \frac{1}{4}$, $x = \sqrt{4 \cdot \frac{1}{4}} = -1$ [positive value does not satisfy the equation of line]

So, the intersection points are $(2, 1)$ and $(-1, \frac{1}{4})$.

$$\text{Therefore, area of shaded region} = \int_{-1}^2 \left(\frac{x+2}{4} \right) dx - \int_{-1}^2 \frac{x^2}{4} dx$$

$$= \frac{1}{4} \left[\frac{x^2}{2} + 2x \right]_{-1}^2 - \frac{1}{4} \left[\frac{x^3}{3} \right]_{-1}^2$$

$$= \frac{1}{4} \left(\frac{4}{2} + 4 - \frac{1}{2} + 2 \right) - \frac{1}{4} \left(\frac{8}{3} + \frac{1}{3} \right)$$

$$= \frac{1}{4} \cdot \frac{15}{2} - \frac{1}{4} \cdot \frac{9}{3}$$

$$= \frac{9}{8} \text{ sq units}$$

26. The area of the region bounded by the curve $y = \sqrt{16-x^2}$ and

x - axis is:

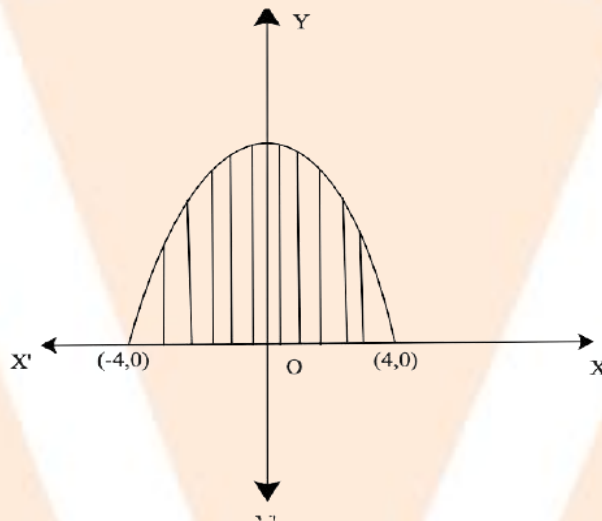
(A) 8π sq. units

(B) 20π sq. units

(C) 16π sq. units

(D) 256π sq. units

Ans: Given equation of curve is $y = \sqrt{16-x^2}$ and the equation of line is x - axis i.e , $y = 0$.



Therefore $\sqrt{16-x^2} = 0$

$$\Rightarrow 16 - x^2 = 0$$

$$\Rightarrow x^2 = 16$$

$$\Rightarrow x = \pm 4$$

So, the intersection points are (4,0) and (-4,0).

Therefore, area of curve, $A = \int_{-4}^4 (16-x^2)^{\frac{1}{2}} dx$

$$= \int_{-4}^4 (4^2 - x^2)^{\frac{1}{2}} dx$$

$$= \left[\frac{x}{2} \sqrt{4^2 - x^2} + \frac{4^2}{2} \sin^{-1} \frac{x}{4} \right]_{-4}^4$$

$$= \left[\frac{4}{2} \sqrt{4^2 - 4^2} + \frac{4^2}{2} \sin^{-1} \frac{4}{4} \right] - \left[\frac{-4}{2} \sqrt{4^2 - (-4)^2} + \frac{4^2}{2} \sin^{-1} \frac{-4}{4} \right]$$

$$= \left[2 \cdot 0 + 8 \cdot \frac{\pi}{2} - 0 + 8 \cdot \frac{\pi}{2} \right]$$

$$= 8\pi \text{ sq units}$$

27. Area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$ is

- (A) 16π sq units (B) 4π sq units (C) 32π sq units (D) 24 sq units

Ans: We have enclosed by x-axis ie. $y = 0$, $y = x$ and the circle $x^2 + y^2 = 32$ in the first quadrant.

$$\text{since, } x^2 + y^2 = 32 \quad [\because y = x]$$

$$\Rightarrow 2x^2 = 32$$

$$\Rightarrow x = \pm 4$$

So, the intersection point of circle $x^2 + y^2 = 32$ and line $y = x$ are (4,4) and (4,-4).

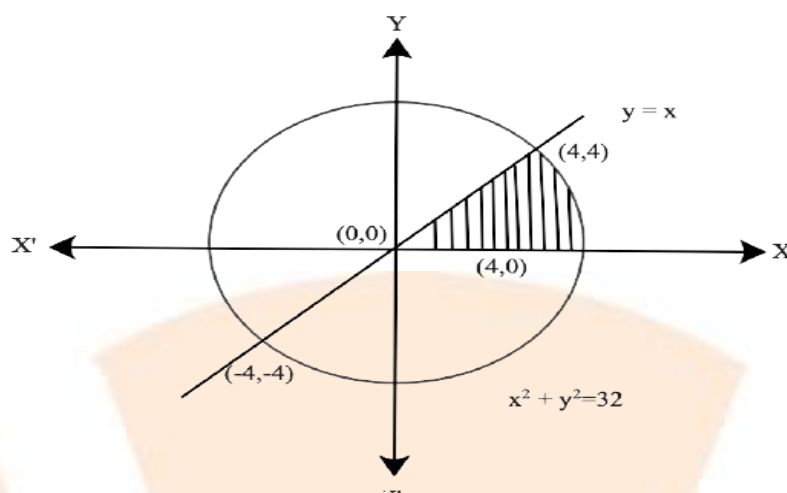
$$\text{And } x^2 + y^2 = (4\sqrt{2})^2$$

Since, $y = 0$,

$$\Rightarrow x^2 + 0 = (4\sqrt{2})^2$$

$$\Rightarrow x = \pm 4\sqrt{2}$$

So. the circle intersects the x-axis at $(\pm 4\sqrt{2}, 0)$

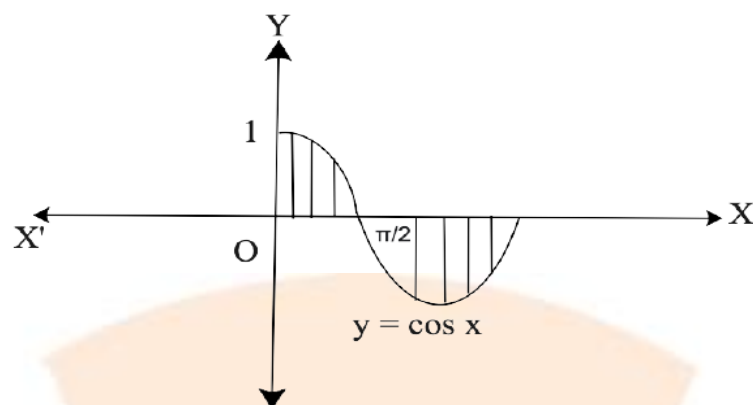


$$\begin{aligned}
 \text{Area of shaded region} &= \int_0^4 x dx + \int_4^{4\sqrt{2}} \sqrt{(4\sqrt{2})^2 - x^2} dx \\
 &= \left[\frac{x^2}{2} \right]_0^4 + \left[\frac{x}{2} \sqrt{(4\sqrt{2})^2 - x^2} + \frac{(4\sqrt{2})^2}{2} \sin^{-1} \frac{x}{4\sqrt{2}} \right]_4^{4\sqrt{2}} \\
 &= \frac{16}{2} + \left[\frac{4\sqrt{2}}{2} \cdot 0 + 16 \sin^{-1} \frac{4\sqrt{2}}{4\sqrt{2}} - \frac{4}{2} \sqrt{(4\sqrt{2})^2 - 16} - 16 \sin^{-1} \frac{4}{4\sqrt{2}} \right] \\
 &= 8 + \left(16 \cdot \frac{\pi}{2} - 2\sqrt{16} - 16 \cdot \frac{\pi}{4} \right) \\
 &= 8 + 8\pi - 8 - 4\pi = 4\pi \text{ sq. units}
 \end{aligned}$$

28. Area of the region bounded by the curve $y = \cos x$ between $x = 0$ and $x = \pi$ is

- (A) 2 sq units
- (B) 4 sq units
- (C) 3 sq units
- (D) 1 sq units

Ans: Required area enclosed by the curve $y = \cos x$, $x = 0$ and $x = \pi$.

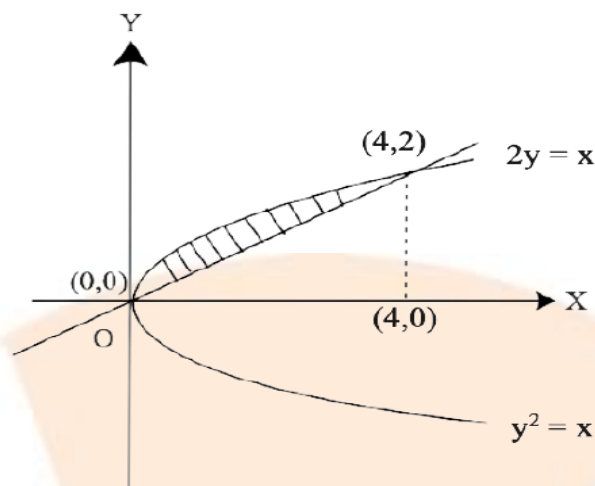


$$\begin{aligned}
 A &= \int_0^{\frac{\pi}{2}} \cos x \, dx + \left| \int_{\frac{\pi}{2}}^{\pi} \cos x \, dx \right| \\
 &= \left[\sin \frac{\pi}{2} - \sin 0 \right] + \left| \sin \frac{\pi}{2} - \sin \pi \right| \\
 &= 1 + 1 = 2 \text{ sq units}
 \end{aligned}$$

29. The area of the region bounded by parabola $y^2 = x$ and the straight line $2y = x$ is

- (A) $\frac{4}{3}$ sq units
- (B) 1 sq units
- (C) $\frac{2}{3}$ sq units
- (D) $\frac{1}{3}$ sq units

Ans: We have to find the area enclosed by parabola $y^2 = x$ and the straight line $2y = x$.



Therefore, $\left(\frac{x}{2}\right)^2 = x$

$$\Rightarrow x^2 = 4x$$

$$\Rightarrow x(x - 4) = 0$$

$$\Rightarrow x = 4 \Rightarrow y = 2 \text{ and } x = 0 \Rightarrow y = 0$$

So, the intersection points are (0,0) and (4,2).

Area enclosed by shaded region,

$$A = \int_0^4 \left[\sqrt{x} - \frac{x}{2} \right] dx$$

$$= \left[\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} - \frac{1}{2} \cdot \frac{x^2}{2} \right]_0^4$$

$$= \left[2 \frac{x^{\frac{3}{2}}}{3} - \frac{x^2}{4} \right]_0^4$$

$$= \frac{2}{3} \cdot \frac{16}{4^{\frac{3}{2}}} - \frac{16}{4} - \frac{2}{3} \cdot 0 + \frac{1}{4} \cdot 0$$

$$= \frac{16}{3} - \frac{16}{4}$$

$$= \frac{64-48}{12} = \frac{16}{12} = \frac{4}{3} \text{ sq units}$$

30. The area of the region bounded by the curve $y = \sin x$ between the ordinates x

$= 0$, $x = \frac{\pi}{2}$ and the x - axis is

(A) 2 sq units

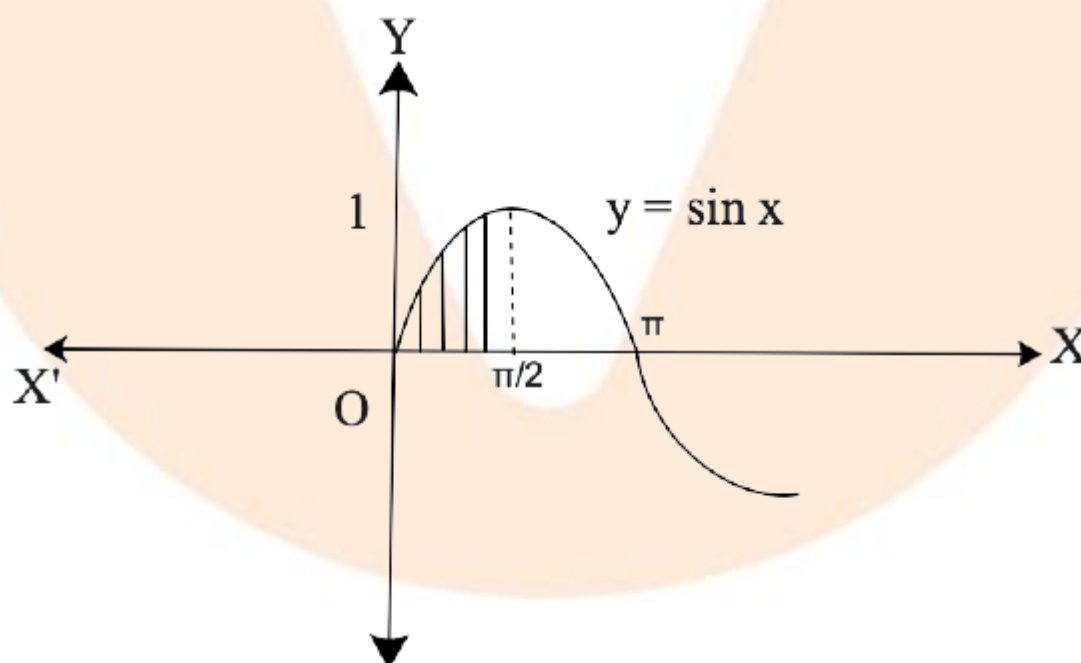
(B) 4 sq units

(C) 3 sq units

(D) 1 sq units

Ans: Area of the region bounded by the curve $y = \sin x$ between between the

ordinates $x = 0$, $x = \frac{\pi}{2}$ and x - axis is



$$A = \int_0^{\frac{\pi}{2}} \sin x \, dx$$

$$= - \left[\cos x \right]_0^{\frac{\pi}{2}} = - \left[\cos \frac{\pi}{2} - \cos 0 \right]$$

$$= -[0-1] = 1 \text{ sq unit}$$

31. The area of the region bounded by the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is

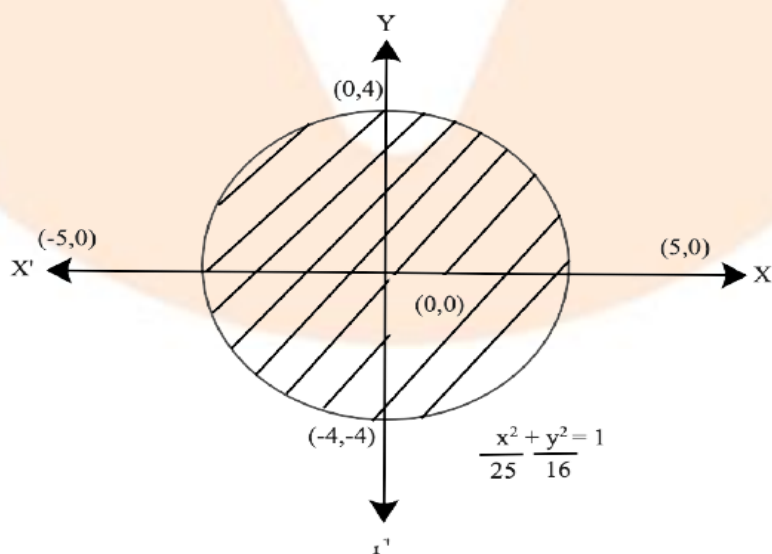
- (A) 20π sq units
 (B) $20 \pi^2$ sq units
 (C) $16 \pi^2$ sq units
 (D) 25π sq units

Ans: We have, $\frac{x^2}{5^2} + \frac{y^2}{4^2} = 1$

Here, $a = \pm 5$ and $b = \pm 4$

And, $\frac{y^2}{4^2} = 1 - \frac{x^2}{5^2}$

$$\Rightarrow y^2 = 16 \left(1 - \frac{x^2}{5^2} \right)$$



$$\Rightarrow y^2 = 16 \left(1 - \frac{x^2}{5^2}\right)$$

$$\Rightarrow y = \sqrt{16 \left(1 - \frac{x^2}{5^2}\right)}$$

$$\Rightarrow y = \frac{4}{5} \sqrt{5^2 - x^2}$$

Area enclosed by ellipse, $A = 2 \cdot \frac{4}{5} \int_{-5}^5 \sqrt{5^2 - x^2} \, dx$

$$= 2 \cdot \frac{8}{5} \int_0^5 \sqrt{5^2 - x^2} \, dx$$

$$= 2 \cdot \frac{8}{5} \left[\frac{x}{2} \sqrt{5^2 - x^2} + \frac{5^2}{2} \sin^{-1} \frac{x}{5} \right]_0^5$$

$$= 2 \cdot \frac{8}{5} \left[\frac{5}{2} \sqrt{5^2 - 5^2} + \frac{5^2}{2} \sin^{-1} \frac{5}{5} - 0 - \frac{25}{2} \cdot 0 \right]$$

$$= 2 \cdot \frac{8}{5} \left[\frac{25}{2} \cdot \frac{\pi}{2} \right]$$

$$= \frac{16}{5} \cdot \frac{25\pi}{4}$$

$$= 20\pi \text{ sq units}$$

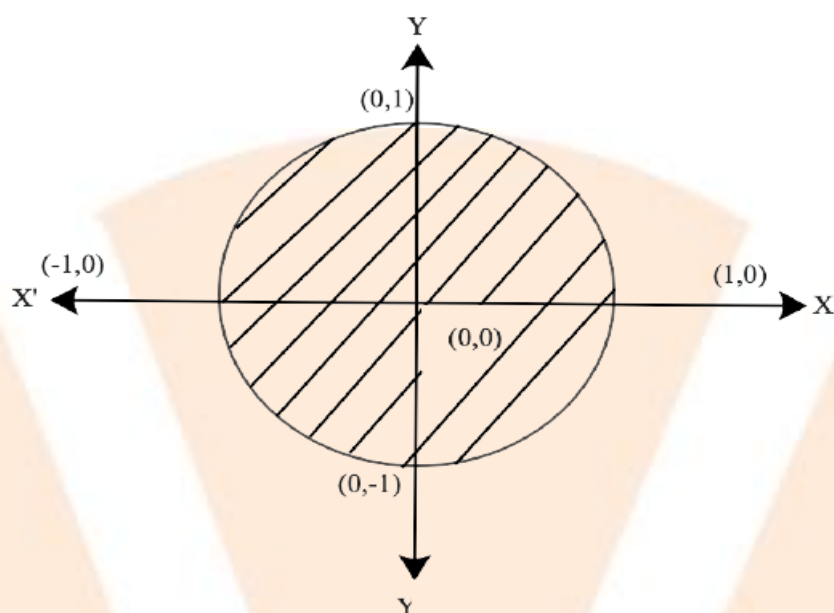
32. The area of the region bounded by the circle $x^2 + y^2 = 1$ is

(A) 2π sq units (B) π sq units (C) 3π sq units (D) 4π sq units

Ans: We have, $x^2 + y^2 = 1$ [Since, $r = \pm 1$]

$$\Rightarrow y^2 = 1 - x^2$$

$$\Rightarrow y = \sqrt{1 - x^2}$$



Therefore, area enclosed by circle = $2 \int_{-1}^1 \sqrt{1^2 - x^2} \, dx$

$$= 2 \cdot 2 \int_0^1 \sqrt{1^2 - x^2} \, dx$$

$$= 2 \cdot 2 \left[\frac{x}{2} \sqrt{1^2 - x^2} + \frac{1^2}{2} \sin^{-1} x \right]_0^1$$

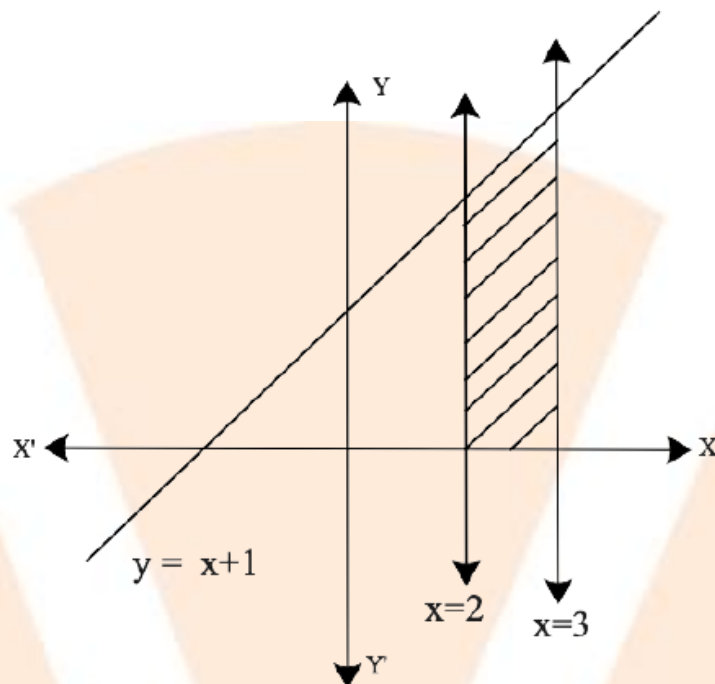
$$= 4 \left[\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{\pi}{2} - 0 - \frac{1}{2} \cdot 0 \right]$$

$$= 4 \cdot \frac{\pi}{4} = \pi \text{ sq units}$$

33. The area of the region bounded by the curve $y = x + 1$ and the lines $x = 2$ and $x = 3$ is

(A) $\frac{7}{2}$ sq units (B) $\frac{9}{2}$ sq units (C) $\frac{11}{2}$ sq units (D) $\frac{13}{2}$ sq units

Ans: Required area , $A = \int_2^3 (x+1) dx$



$$\Rightarrow \left[\frac{x^2}{2} + x \right]_2^3$$

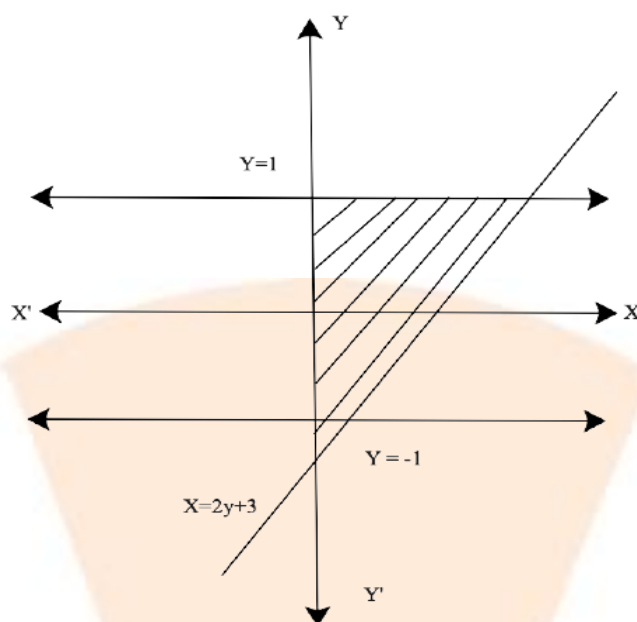
$$\Rightarrow \left[\frac{9}{2} + 3 - \frac{4}{2} - 2 \right]$$

$$\Rightarrow \frac{7}{2} \text{ sq units}$$

34. The area of the region bounded by the curve $x = 2y + 3$ and the y lines. $y = 1$ and $y = -1$ is

(A) 4 sq units (B) $\frac{3}{2}$ sq units (C) 6 sq units (D) 8 sq units

Ans: Required area , $A = \int_{-1}^1 (2y+3) dy$



$$\Rightarrow \left[\frac{2y^2}{2} + 3y \right]_{-1}^1$$

$$\Rightarrow [y^2 + 3y]_{-1}^1$$

$$\Rightarrow [1+3-1+3]$$

$$\Rightarrow 6 \text{ sq. units}$$